Developing A Domain-Specific LLM for Optical Networks: A Reinforcement Learning-Based Fine-Tuning Framework

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***Abstract*—Optical networks serve as the backbone of modern communication infrastructure, where efficient operation and maintenance (O&M) are essential for ensuring reliable and high-speed data services. However, traditional network O&M face persistent challenges, including high labor costs, delayed response time, and difficulties in processing massive and complex network data. Although large language models (LLMs) have demonstrated strong capabilities in text understanding, generation, and reasoning, their direct application in optical network O&M is limited by domain-specific knowledge barriers, inherent reasoning biases, and insufficient performance in complex multi-step tasks. To address this issue, this study develops a domain-adaptation and system-implementation framework that applies two established reinforcement learning-based fine-tuning methods (RLHF and ReFT) to construct domain-specialized LLMs for optical network O&M tasks. In the context of log analysis, RLHF achieves improvements of 1.64 points in accuracy, 1.02 points in content richness, and a notable 10-point increase in interactivity over supervised fine-tuning. In alarm localization, ReFT achieves accuracy improvements of 2%–13% across four reasoning tasks. The extensive tests not only demonstrate the practical value of RL-based fine-tuning in enhancing alignment and reasoning for domain-specific applications, but also provides a practical methodology and implementation reference for applying reinforcement learning-based LLM adaptation in optical network O&M environments.**

***Index Terms*—Large language model, reinforcement learning from human feedback, reinforced fine-tuning, optical networks.**

I. Introduction

Optical networks constitute the backbone of modern communication infrastructure, where efficient operation and maintenance (O&M) play a critical role in ensuring high-speed data transmission, reliable service delivery, and secure network environments [1]. However, traditional optical network O&M paradigms still rely heavily on human expertise and manual operations, leading to high labor costs, complex data analysis processes, and delayed fault response [2], [3], [4]. As optical networks continue to scale in size and complexity, these limitations increasingly hinder the realization of intelligent, autonomous, and efficient network management.

In recent years, the rapid development of generative artificial intelligence (GenAI) has brought large language models (LLMs), such as ChatGPT [5] and LLaMA [6], [7], to the forefront of artificial intelligence research. Benefiting from large-scale pretraining, LLMs exhibit strong capabilities in natural language understanding, text generation, and complex reasoning [8]. These models have demonstrated remarkable success across a wide range of real-world applications. For example, in code review tasks, LLMs are capable of automatically generating review comments, assisting engineers in understanding code logic, identifying potential defects, and suggesting revisions, thereby improving development efficiency and code quality [9]. In network intrusion detection, LLMs have been leveraged to analyze traffic patterns and abnormal behaviors, infer attack paths, and generate interpretable security reports and response strategies, enhancing both detection accuracy and system explainability [10]. These achievements highlight the strong generalization ability and task adaptability of LLMs, suggesting their potential to support intelligent O&M in optical networks.

Motivated by these successes, researchers have begun to explore the application of LLMs to optical network O&M tasks, including quality-of-transmission (QoT) estimation and optimization [11], simulation assistance [12], and automated network control [13], [14], [15], [16]. Despite their promising performance, directly deploying general-purpose LLMs in optical network scenarios remains challenging. Optical networks are highly specialized systems that demand extreme robustness, reliability, and domain-specific reasoning. In practice, LLMs often suffer from domain knowledge gaps [17], inherent reasoning biases [18], and task-specific constraints [19], which limit their effectiveness in professional O&M applications. These challenges underscore the necessity of adapting LLMs to the optical network domain.

To bridge the gap between general-purpose LLMs and domain-specific optical network tasks, model adaptation techniques are required. Among existing approaches, reinforcement learning (RL)-based fine-tuning has emerged as a particularly promising direction. Unlike conventional super-

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vised learning, RL-based methods enable models to optimize their behavior through interaction with reward signals, allowing them to go beyond simple imitation of labeled data. By incorporating rule-based or human preference-based feedback, RL-based fine-tuning can enhance alignment with expert intent, reduce hallucinations, and improve reasoning consistency. These properties are especially desirable for complex optical network O&M tasks, such as log analysis and alarm root cause localization, which involve multi-step reasoning and high interpretability requirements.

In this paper, RL-based fine-tuning techniques are applied to address complex and professional tasks in optical networks, and a dual-strategy adaptive framework is constructed to support LLM-based network log analysis and alarm localization. The framework leverages established reinforcement learning approaches to enhance the domain adaptability of LLMs for optical network operation and maintenance tasks. Through empirical analysis and validation on two representative case studies, this work demonstrates how reinforcement learning can be effectively integrated into LLMs in optical network scenarios and provides practical insights for applying RL-based methods in optical networks and related domains.

The main contributions of this work include:

1) The implementation processes and applicable scenarios of typical RL techniques in LLMs are summarized, providing practical guidance for domain specialization.
2) RLHF is applied to enhance the alignment between optical network log analysis outputs and human intent. Task accuracy of 9.09 points, richness of 7.2 points, and interactivity of 10 points are achieved by the final model. Compared with the SFT model, the accuracy of the preference model is improved by 1.64 points, richness is increased by 1.02 points, and interactivity is increased by 10 points.
3) ReFT is employed to improve LLM performance in complex reasoning tasks related to network RCL. The results show that accuracy in all four tasks was improved: the task of extracting topological relationships of real-time alarms was completed with 98% accuracy, an increase of 2%; the task of extracting alarm propagation of real-time alarms was completed with 99% accuracy, an increase of 13%; the generation of the alarm event matrix was achieved with 99% accuracy, an increase of 5%; and the identification of root alarms was achieved with 96% accuracy, an increase of 6%.
4) An in-depth analysis of the validation outcomes is conducted, and key challenges and practical insights are summarized, providing valuable guidance for the application of RL-based fine-tuning to LLM.

The remainder of this paper is organized as follows. Section II reviews related work on LLM applications in optical network O&M and existing domain adaptation techniques, with a focus on supervised and reinforcement learning-based fine-tuning methods. Section III presents the principles and overall framework of the proposed dual-strategy reinforcement learning adaptation approach, including RLHF and ReFT. Section IV describes the two representative tasks considered in this study, namely log analysis and alarm root cause localization, along with their corresponding reinforcement learning strategies. Section V details the implementation settings, datasets, hyperparameter configurations, and evaluation metrics. Section VI provides performance evaluation and performance analysis for both tasks. Finally, Section VII concludes the paper and discusses future research directions.

II. Relate Work

A. LLM Applications in Optical Network O&M

With the increasing maturity of large language models, several studies have investigated their application to optical network O&M tasks. Existing works have explored the use of LLMs for quality-of-transmission estimation and optimization [11], network simulation assistance [12], and automated control and decision-making [13], [14], [15], [16]. These studies demonstrate that LLMs can effectively leverage textual and structured information to support network management tasks. However, most existing approaches rely on directly applying pretrained or lightly adapted models, which often struggle with domain-specific constraints, complex operational rules, and high reliability requirements intrinsic to optical networks.

B. Domain Adaptation Techniques for LLMs

To improve LLM performance in specialized domains, three primary adaptation strategies have been widely studied: external augmentation, prompt engineering, and model fine-tuning [20], [21], [22]. External augmentation enables LLMs to access external knowledge bases or tools, offering flexibility and extensibility. However, its effectiveness strongly depends on retrieval accuracy and knowledge coverage, and frequent external queries may introduce significant latency [23]. Prompt engineering aims to elicit task-specific behavior by carefully designing instructions and examples. While effective for simple tasks, this approach often exhibits limited reasoning depth and requires long input prompts, which are constrained by the model's context window [24]. This limitation is particularly pronounced in optical network tasks that involve detailed operational rules, network topology, and resource allocation strategies.

Model fine-tuning adjusts the internal parameters of LLMs to better capture domain-specific patterns and reasoning processes. Compared with external augmentation and prompt engineering, fine-tuning generally achieves stronger domain adaptation, albeit at the cost of additional training effort and computational resources.

C. Supervised and Reinforcement Learning-Based Fine-Tuning

Among fine-tuning approaches, supervised fine-tuning (SFT) has been widely adopted due to its simplicity and effectiveness. By learning from expert-labeled data, SFT can significantly improve LLM performance on specific tasks [25]. In optical network O&M, SFT has been applied to tasks such as log analysis and alarm root cause localization [26], [27]. However, existing studies reveal notable limitations. SFT-trained models often fail to fully capture expert

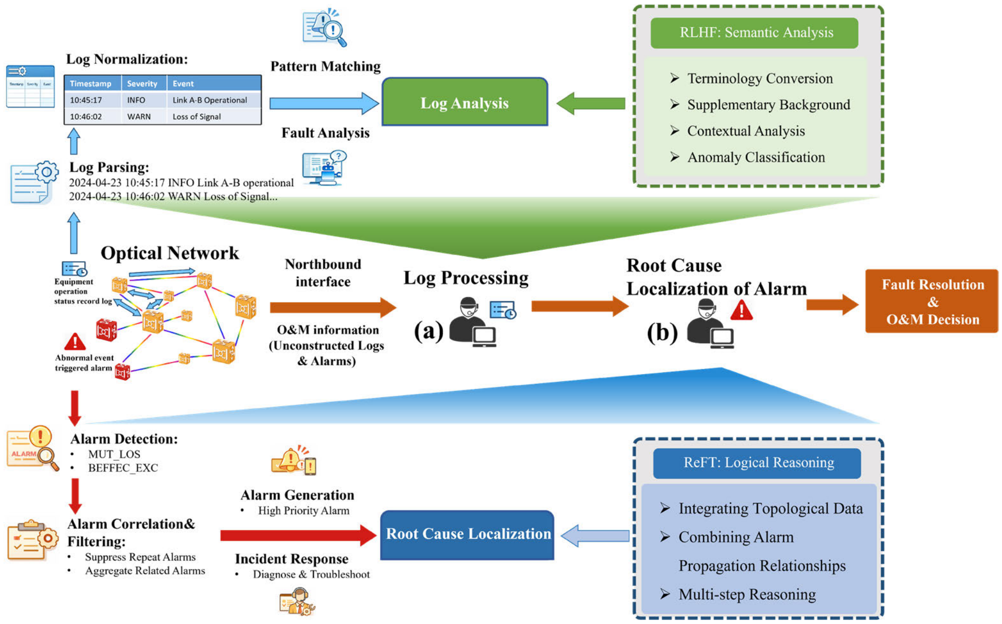

Fig. 1. Optical network operation and maintenance process based on (a) logs and (b) alarms analysis under dual strategies.

intent, leading to incomplete analyses, hallucinated network parameters, or missing recommendations. Moreover, their reasoning accuracy remains insufficient for complex multi-step tasks.

To address these limitations, reinforcement learning-based fine-tuning has gained increasing attention [28], [29]. Reinforcement Learning from Human Feedback (RLHF) utilizes a reward model trained on human preferences to guide policy updates, improving alignment with human intent and reducing hallucinations [30]. Reinforced Fine-Tuning (ReFT), in contrast, directly leverages environment feedback without preference-labeled data, demonstrating strong improvements in multi-step reasoning and decision-making capabilities [31]. These advances indicate that RL-based fine-tuning offers a powerful mechanism for enhancing alignment, robustness, and reasoning performance of LLMs in domain-specific applications.

III. Principle and Framework

In optical network O&M, two representative and tightly coupled tasks are considered in this work: log analysis and alarm root cause localization.

Log analysis focuses on interpreting raw operational logs generated by heterogeneous network devices, with the goal of resolving ambiguous semantics, normalizing domain-specific terminology, and enriching contextual information related to network states and events. Alarm root cause localization, on the other hand, aims to identify the underlying faults responsible for observed alarm patterns by reasoning over network topology and alarm propagation relationships.

In practical O&M scenarios, these two tasks are not performed independently but are naturally organized as a task-level workflow. Log analysis typically serves as the entry point of the O&M process, where raw logs are first transformed into structured and semantically clarified representations. This process provides a coherent understanding of network behavior and supports preliminary anomaly identification, as illustrated in Fig. 1(a).

Building upon the clarified semantics obtained from log analysis, alarm root cause localization further integrates network topology information and alarm propagation dependencies to infer potential fault propagation paths. As a result, alarm localization inherently relies on the outputs of log analysis, forming a sequential O&M workflow in which higher-level reasoning and decision-making are grounded in prior semantic understanding rather than performed in isolation, as shown in Fig. 1(b).

To adapt LLMs to this continuous and dependency-aware operational workflow, RL–based fine-tuning is introduced at different stages of the pipeline using two established techniques: RLHF and ReFT. RLHF is applied at the log analysis stage to enhance domain semantic alignment, operational preference modeling, and context-aware interpretation of raw logs, enabling the generation of structured and semantically consistent representations (Fig. 1(a)). Subsequently, ReFT is employed at the alarm localization stage to improve inference consistency and multi-step logical reasoning for topology-aware analysis and alarm propagation inference. Although RLHF and ReFT target different optimization objectives and are implemented independently, they together form a logically ordered adaptive strategy that reflects the inherent sequence of optical network O&M tasks.

From an implementation perspective, both RLHF and ReFT follow a unified reinforcement learning based fine-tuning paradigm. Specifically, they share a common three-stage training pipeline consisting of supervised fine-tuning (SFT), reward model construction, and policy optimization, as shown in Fig. 2. In the remainder of this section, we provide a detailed description of the technical principles and implementation details of these two methods within the proposed framework.

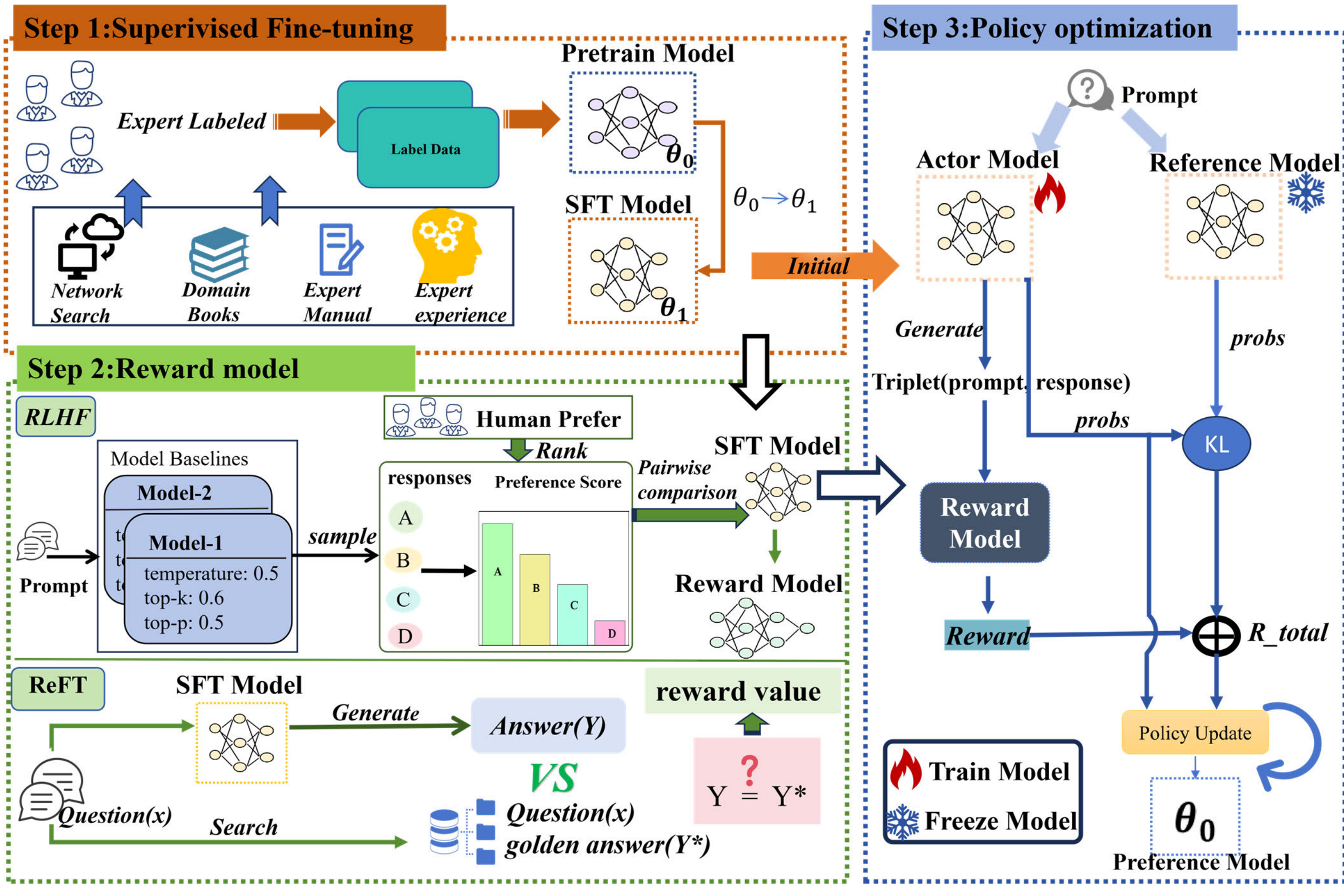

Fig. 2. Three-stage implementation process of RLHF and ReFT: The framework consists of three sequential stages. (a) Supervised Fine-Tuning: Fine-tunes the base model on labeled data. (b) Reward Model: In RLHF, a reward model is trained based on human preference data to score model outputs, while in ReFT, a handcrafted reward function is designed to directly evaluate responses. (c) Policy Optimization: Optimizes the model policy via reinforcement learning, guided by the reward model or reward function.

A. Supervised Fine-Tuning

SFT refers to the process of fine-tuning a pre-trained LLM on a labeled dataset tailored to specific downstream tasks, thereby enhancing the model's adaptability to solve related problems [32], as shown in Fig. 2 (a). In this context, we consider an LLM parameterized by θ. Let x and y denote the model's input and the corresponding output label, thereby forming the SFT dataset $D_{SFT} = \{x_i, y_i\}_{i=1}^{N}$, where N represents the number of samples in the SFT dataset. In theory, the goal of SFT is to adjust the model parameters so that the model generates a response similar to the label with a higher probability. Therefore, the training objective of SFT can be formally expressed as [33]:

$$L_{SFT}(\theta) = -E_{x,y\sim D_{SFT}}[\log p_\theta(y|x)] \tag{1}$$

where $p_\theta(y|x)$ represents the model's predicted probability of the correct label y given the input x.

B. Reward Model

The reward model plays a pivotal role in providing feedback on the actions of the model in RL algorithms, as shown in Fig. 2(b). In RLHF, an explicit reward model is trained based on human labelers' preferences, enabling the model to assign higher rewards to responses that better align with human expectations. In ReFT, although no explicit reward model is trained, a task-specific reward function or evaluation mechanism is constructed to assess whether the model's outputs match predefined correct answers in mathematical reasoning tasks. In this sense, both RLHF and ReFT involve forms of task-oriented modeling to guide fine-tuning, ensuring that the model's behavior aligns with either human preferences or established task standards.

1) Reward Modeling of RLHF: In RLHF, the first step in training a reward model is to collect preference data from a group of human labelers. Different responses to the same questions are then sampled from various model baselines [34] and presented to the human labelers for ranking. The resulting data is transformed into a set of prompt-chosen-rejected trios and can be denoted as $D_{Reward} = \{x^{(i)}, y^{(i)}_{\text{chosen}}, y^{(i)}_{rejected}\}_{i=1}^{N_1}$, where the chosen response is preferred over the rejected response for training and the N_1 is the number of samples in the reward dataset.

In theory, training the reward model is based on the Bradley-Terry model [35], which is used to estimate the probability that pairwise comparison $i < j$ is true, as calculated:

$$\mathrm{P\,(i > j)} = \frac{e^{\beta_i}}{e^{\beta_i} + e^{\beta_j}} \tag{2}$$

where e^{β_i} and e^{β_j} represent the probabilities assigned to individuals i and j respectively.

The reward model should assign a higher score to the preferred response than to the rejected one. Therefore, the objective function of training the reward model is:

$$L_{Reward}(\theta) = -E_{(x, y_{chosen}, y_{rejected}) \sim D_{Reward}} [\log(\sigma(r_\theta(x, y_{chosen}) - r_\theta(x, y_{rejected})))] \quad (3)$$

In practice, the reward model is typically implemented either by appending a linear layer to predict a single logit or by removing the final decoding layers and replacing them with a linear layer.

2) Reward Function of ReFT: In the reward function of ReFT, the reward values indicate whether the final answer is correct. Accordingly, the reward function is designed to compare model-generated answers with the corresponding golden answers. To achieve this, it is essential to construct a dataset consisting of (question, golden answer) pairs, denoted as (x, y). When generating outputs for the corresponding inputs x in the model, the answers y^* are extracted and compared with the golden answers y, based on the dataset's predefined format.

C. Proximal Policy Optimization

Proximal Policy Optimization (PPO) is a typical off-policy Actor-critic RL algorithm [36]. A common approach is to choose the SFT model as the initial policy in PPO, denoted as π_{SFT} (y|x). A new dataset only concludes prompt needs to be prepared as $D_{PPO} = \{x_i\}_{i=1}^{N_2}$, among them, N_2 represents the number of datasets in PPO. In the process of training, a batch of data is fed into the Actor model, which generates a sequence of input-output pairs (x, y). The reward model then assigns a reward value r (x, y) to each pair. Subsequently, optimization iterations are carried out with the aim of obtaining higher reward values. Therefore, the objective of PPO training is to maximize the following objective function:

$$L(\pi_\phi^{RL}) = -E_{x \sim D_{RL}, y \sim \pi^{RL}(y|x)}[r(x, y) - \lambda \cdot D_{KL}(\pi_\phi^{RL}(y|x) || \pi^{SFT}(y|x))] \quad (4)$$

where $\pi_\phi^{RL}(y|x)$ is the learned RL policy with parameters φ, the Kullback-Leibler (KL) regularization [37] is to prevent the new model from deviating too far from the original initial model, to avoid mode collapse, $\beta \succ 0$ is the regularization parameter is to control the deviation of the new model from the reference model, as shown in Fig. 2 (c).

Among Equation (4) is intended to present a conceptual formulation of the PPO optimization objective, rather than a full description of all implementation details.

To explicitly characterize the optimization process, the PPO objective can be expressed in its clipped surrogate form as

$$L^{\text{clip}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right] \quad (5)$$

where the probability ratio between the updated policy and the reference policy is defined as

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}. \quad (6)$$

The term $\hat{A}_t$ denotes the advantage estimate, which is introduced to reduce variance in policy gradient estimation. A value-function baseline $V_\phi(s_t)$ is employed, yielding

$$A_t = Q_t - V_\phi(s_t). \quad (7)$$

In practice, the advantage is computed using Generalized Advantage Estimation (GAE), where the temporal-difference residual is defined as

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t), \quad (8)$$

and the resulting advantage estimate is

$$\hat{A}_t = \sum l = 0^\infty (\gamma\lambda)^l \delta_{t+l}. \quad (9)$$

For numerical stability, the advantages are normalized within each mini-batch before being substituted into the clipped objective, thereby completing the policy update procedure

In summary, RLHF is particularly effective in capturing implicit human preferences by learning a reward model from human feedback, which serves as a proxy for subjective quality. This model guides policy optimization to improve naturalness and alignment with human intent. In contrast, ReFT directly defines reward functions based on task-specific rules or correctness criteria, offering greater transparency and control. While RLHF learns the reward signal from data, ReFT designs it explicitly—highlighting a key difference in their underlying principles and making them suitable for different types of tasks.

IV. Tasks Description

In this section, we evaluate the effectiveness of the dual-strateg adaptation framework by studying two representative tasks in optical networks: optical network log analysis and alarm root cause localization.

A. Log Analysis Task Based on RLHF

Task Requirements:: optical network logs are valuable data resources generated by network devices and links during operation, recording network activities, events, status changes, and abnormal conditions. The logs can be used to track network behavior, monitor operational status, and ensure system security [38]. Due to their high-frequency generation, log analysis is significant for network status monitoring, fault diagnosis, and anomaly detection, supporting daily network O&M. Therefore, fully leveraging optical network logs is essential for supporting daily O&M, ensuring network stability, and enhancing operational efficiency. In our previous work [26], anomaly analysis of logs was achieved through supervised fine-tuning of the LLaMA-2-7B-chat model using optical network log data, which effectively explained key information in anomaly event descriptions and generated actionable insights. The model's performance was evaluated across three critical dimensions: correctness, usability, and fluency. However, our analysis revealed a significant limitation – the generated content frequently exhibited inaccuracies that deviated from the original log information. This highlights the need for more effective fine-tuning techniques.

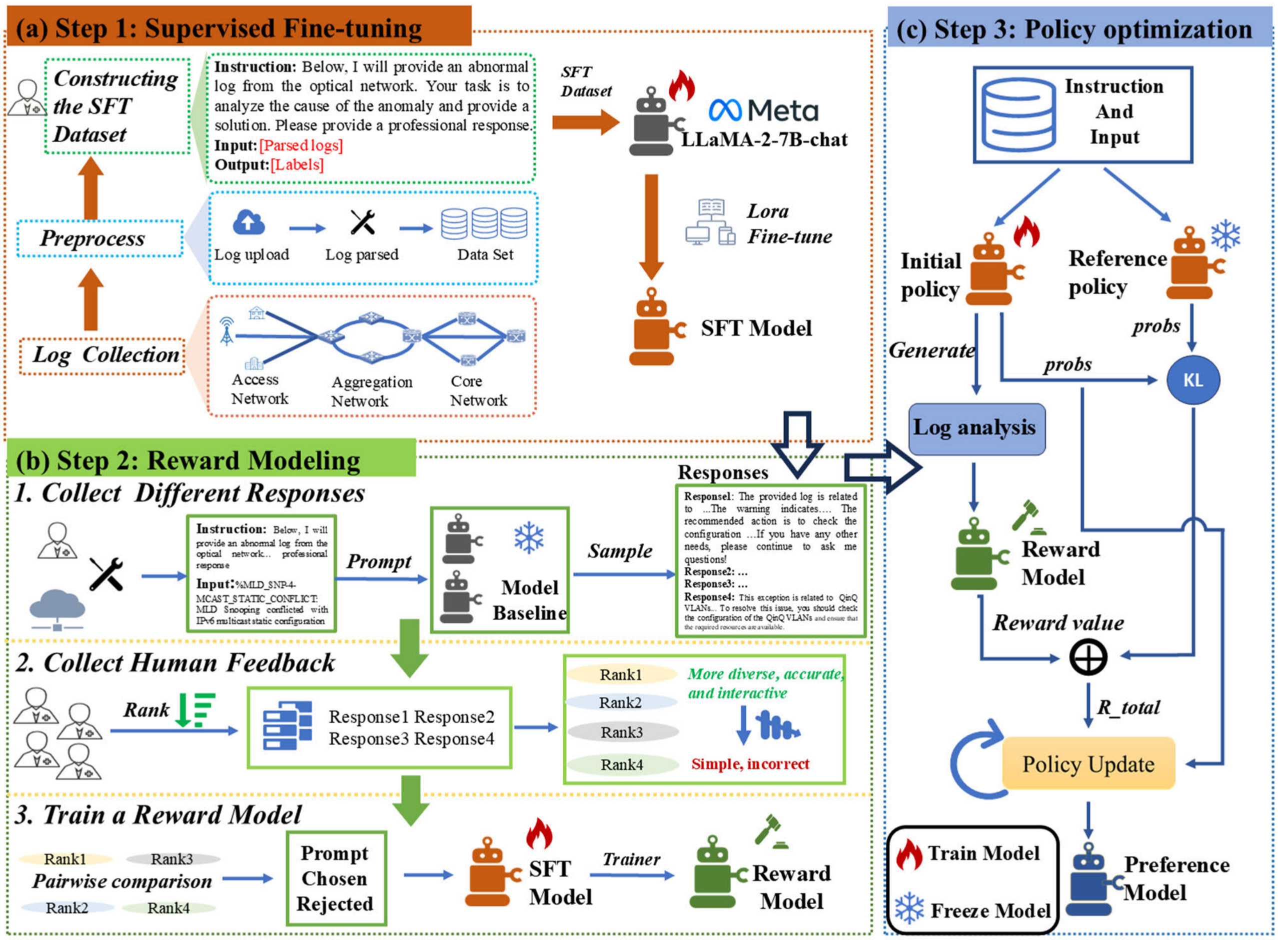

Fig. 3. The implementation process of the log analysis task based on RLHF consists of three stages. (a) Supervised Fine-Tuning: The base model is fine-tuned on labeled log data to learn domain-specific patterns and anomaly detection capabilities. (b) Reward Modeling: Human feedback is collected by ranking model-generated log analysis results, and a reward model is trained to predict these preference scores. (c) Policy Optimization: Using the PPO algorithm, the model policy is optimized with guidance from the reward model, encouraging it to generate more accurate, informative, and human-aligned log analysis outputs.

To address this issue, the RLHF is implemented to enhance output precision. In addition, we further investigate LLM's application-specific requirements in optical network O&M scenarios. In general, the effectiveness of log analysis is directly correlated with the comprehensiveness and level of detail in its outputs. Therefore, we introduce additional preference designs, firstly emphasizing two key aspects of content richness: (1) clear and accessible analysis of log background, sources, and content; (2) in-depth, multi-dimensional recommendations for abnormal log resolution. Secondly, to enhance human-machine interaction for network operators, the system's original direct expression style is transformed into a more dynamic and flexible narrative form, thereby improving its practicality in real-world operational scenarios. The implementation process of the verification comprises three main steps, as shown in Fig. 3. In the first step, the model is fine-tuned on log-labeled data via SFT. Next, multiple SFT models are used to collect human preference data, based on which a reward model is trained. In the final step, the reward model guides the optimization of the SFT model through RL, resulting in a preference-aligned model.

Human preference dataset: the first step in reward modeling is to construct a high-quality dataset of human preferences, as shown in Fig. 3 (b). For the task requirements of log analysis, we adopt the following process to collect human preference datasets.

Firstly, we randomly sampled a subset of the SFT dataset and processed it using various model baselines, which correspond to the different epochs of SFT training. In addition, when the model generates sampling replies, we employ the Diverse Beam Search decoding method [39] with a beam budget set to 4 and repeatability and diversity penalty parameters both set to 1.5. This approach ensures diverse and high-quality responses, and the higher the quality of the preference dataset, the more advantageous it is for improving the accuracy of the trained reward model.

Secondly, the collected responses were evaluated by domain experts with the Zhongjin [40] annotation tool, following a set of a unified set of criteria: accuracy, richness, and interactivity. Manual improvements were applied to enhance its interactivity, making the output more engaging and informative.

Thirdly, for each log analysis instance, only the top four ranked responses were retained, with richness gradually decreasing from the first to the fourth. The top three responses were required to maintain both accuracy and interactivity, whereas the fourth response was intentionally left with potential inaccuracies while retaining its original style. This design choice was made to ensure that the constructed preference pairs (chosen, rejected) differ clearly in at least one evaluation metric, strengthening the consistency and reliability of the preference signal.

By structuring the dataset in this way, this methodology provides a robust foundation for training reward models, emphasizing discernible differences in response quality. Such an approach enhances the model's ability to learn fine-grained preferences, ultimately enhancing performance on log-analysis tasks.

Model selection: during the SFT stage, to ensure continuity, the same base model (LLaMA-2-7b-chat) is selected for study. Subsequently, the trained LLM is then used as the initial policy for the PPO.

For the reward model, instead of initializing it directly from the base model, we choose to initialize it using an SFT model during the reward modeling stage. Furthermore, to evaluate the performance of the reward model, we consider two dimensions: 1) the static discriminative ability of the reward model, which is mainly measured by calculating the preference difference between the 'chosen' and 'rejected' responses in the test set, that is $\mathrm{r}(x, y_{chosen}) - r(x, y_{rejected})$. Specifically, it includes two aspects: a) accuracy, the proportion of samples with positive statistical reward differences. A higher proportion indicates better accuracy in distinguishing good from poor samples. b) Distinguishing strength, calculating the mean of reward differences for all samples. The larger the mean, the stronger the sensitivity and discriminative power of the model to quality differences [41]. 2) The dynamic guidance ability of the reward model, which mainly evaluates the actual effectiveness of the reward model in guiding policy optimization in PPO training, as shown in Fig. 3 (c).

B. Alarm Root Cause Localization Task Based on ReFT

Task requirements: in optical network O&M, alarms serve as essential mechanisms for detecting and indicating network faults or abnormalities. As network scale and complexity grow, the volume of alarms generated during failures increases significantly, with each device or link potentially triggering alarms to signal potential issues and prompt operator intervention. The challenge lies not only in handling large volumes of alarms but also in accurately identifying the true root-cause alarms. Timely and precise RCL is critical for maintaining network stability and ensuring continuous, reliable service delivery. Based on the previous work [27], we collected the real optical transport network (OTN) alarm data and leveraged the relationship between network topology and alarm events to fine-tune the LLM to achieve alarm root cause localization. We divide the implementation into four steps based on the principle of root cause localization. The first step is to generate the alarm event network topology adjacency matrix, which needs to extract the network elements appearing in the real-time alarm sequence from the inherent network topology adjacency matrix of physical links. The second step is to generate the alarm event alarm propagation relationship adjacency matrix, which needs to extract the alarm events appearing in the real-time alarm sequence from the inherent alarm events. The third step is to use the matrices obtained above to construct the alarm event propagation matrix, and the last step is to use the alarm event propagation matrix to convert it into the propagation relationship between the network element and the alarm to determine the root alarm.

In our previous study, we fine-tuned four LLMs using CoT data, with each model focusing on a specific step of the alarm analysis process. To further improve task accuracy, the ReFT technique is introduced in this work. The sequential workflow of the verification is illustrated in Fig. 4.

RCL of alarm CoT data construction. Building a suitable CoT dataset can maximize the understanding of the model and the learning of the algorithm. By utilizing the powerful generation capability of LLM (GPT-4o) to generate CoT data, the performance of small models (LLaMA-3-8B-Instruct) in complex reasoning and step-by-step thinking can be fully stimulated. Therefore, when generating datasets for each task, as shown in Fig. 4 (a), experts first guide GPT-4o to automatically generate CoT training data containing detailed inference processes through fast engineering methods, and then use topology information and alarm sequence information in the real network to design appropriate instructions and input information for each task, expanding the CoT data into a training dataset in the form of {instruction, input, output} to fine tune the small model (LLaMA-3-8B-Instruct).

Rule-based reward function: due to the relatively fixed output format designed for each task in this study, it is often only necessary to verify whether the result in the last inference step of generating the response is consistent with the expected result when verifying whether the output response is consistent with the expected result.

Based on this feature, this study designed a rule-based reward function to systematically evaluate the correctness of generated responses. According to an analysis of outputs produced by the alarm SFT baseline model, the generated responses can be categorized into three types: (1) correct alarm analysis obtained through reasonable logical reasoning, (2) incorrect alarm analysis caused by flawed reasoning, and (3) responses in which the reasoning process or final analysis is incomplete. Correspondingly, reward values of 1.0, 0.1, and 0.0 are assigned to these three cases, respectively, as shown in Fig. 4(b). This hierarchical reward mechanism refines the evaluation criteria for generation quality, strengthens the model's ability to generate correct responses, and enhances robustness [42]. In addition, the asymmetric non-negative reward design was adopted based on empirical observations of training stability. Compared with symmetric reward structures with negative penalties, which tend to produce predominantly negative advantages during early PPO training due to the limited initial accuracy of the SFT model, the proposed reward scheme provides denser learning signals and leads to more stable policy updates and improved training efficiency.

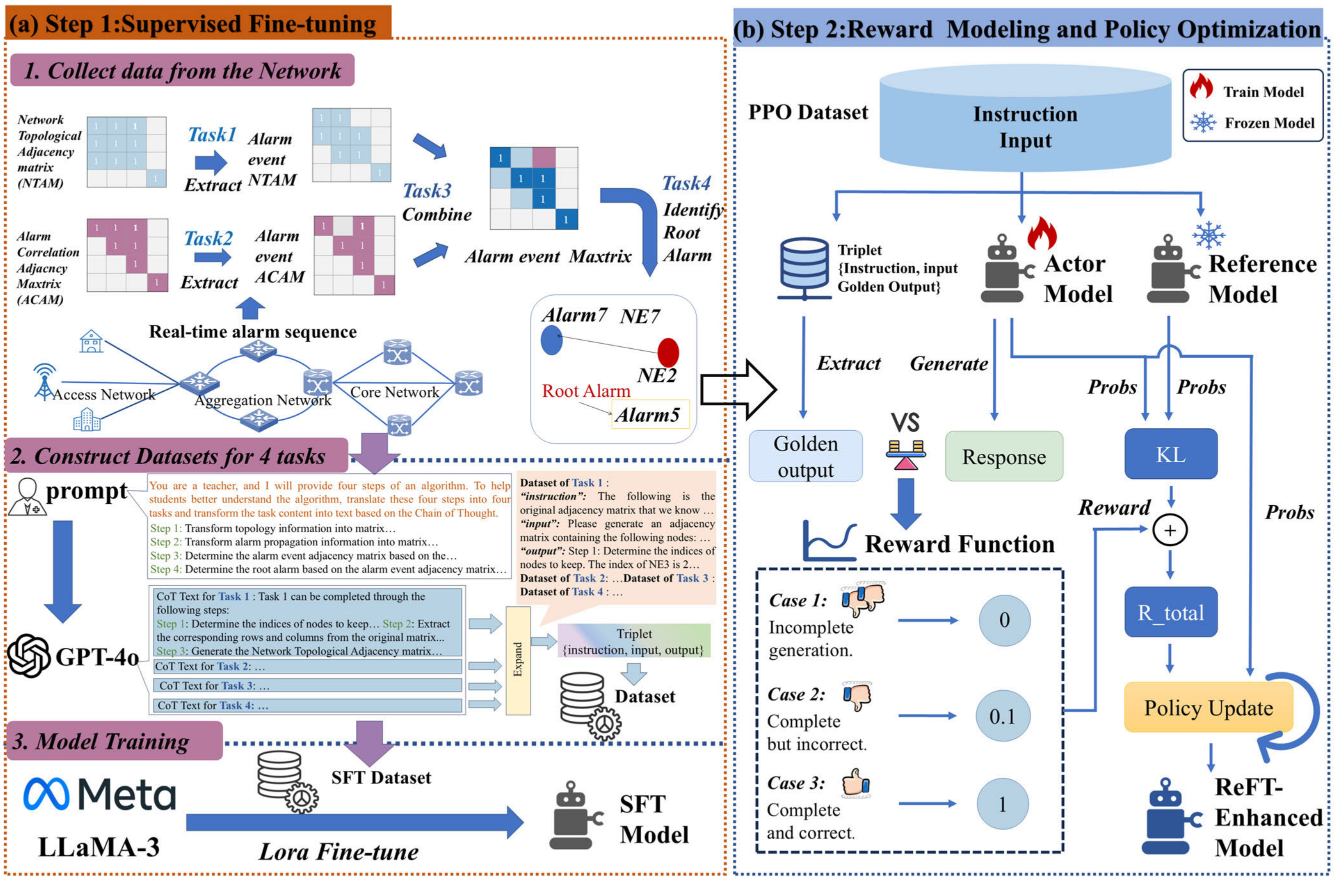

Fig. 4. The implementation process of the root alarm localization task based on ReFT involves two stages. (a) Supervised Fine-Tuning: The base language model is fine-tuned on labeled alarm data to capture domain-specific patterns. (b) Reward Modeling and Policy Optimization: Custom reward functions are designed based on response feedback to evaluate the model's outputs, and the model policy is then optimized using the PPO algorithm guided by these reward functions to improve localization performance.

V. Implementation Details and Evaluation Metrics

A. Log Analysis Task

1) Dataset: The dataset of network logs is collected from a practical telecommunication network, which is composed of both switch logs and OTN logs provided by the operator from 2023.10.31 to 2023.11.2. The number and content of switch logs are rich and easy to obtain. Its addition provides more abundant and effective training data, and its similar format and syntax features help in analyzing optical network logs. Then, the collected logs are annotated by domain experts, who refer to technical manuals and other references to provide labels for log parsing, anomaly detection and classification, and anomaly analysis. The dataset used for log parsing, anomaly detection, and classification consists of 4,975 records.

Since anomaly analysis aims at event-level reasoning rather than long-term temporal trend modeling, one sample per abnormal log template was retained to construct the fine-tuning dataset for anomaly analysis. This approach ensures coverage of diverse anomaly types and semantic patterns while avoiding redundant normal-operation logs. The resulting dataset consists of 609 records, which is divided into a 90% training set and a 10% testing set in the SFT stage. The short-term, high-density nature of the dataset is conducive to capturing the structural distribution and semantic characteristics of abnormal events for model fine-tuning.

In addition, due to the stronger openness, subjectivity, and diverse output space of log analysis tasks, RLHF can effectively optimize the response quality of such open tasks. However, log parsing and anomaly detection are high-certainty tasks, and the benefits brought by RLHF are limited. Therefore, in this paper, we mainly focus on the study of log analysis tasks.

In the reward modeling stage, the reward dataset comprises 1,800 comparison pairs. During training, 1,500 pairs were used as the training set, while the remaining 300 pairs were reserved for testing. Additionally, we collected 3,505 unlabeled data samples from the optical network to construct the PPO dataset, which does not require expert annotation and facilitates optimization iterations.

From a token-level perspective, the datasets used in the SFT, reward modeling, and PPO stages exhibit moderate and well-controlled sequence lengths. In the SFT stage, each sample consists of a fixed instruction (37 tokens), a templated log input (7–72 tokens), and a fault analysis output (19–343 tokens), totaling approximately 66k training tokens and 6.3k test tokens. The reward modeling stage includes 1,800 preference pairs with broader response-length distributions, resulting in approximately 539k training tokens and 107k test tokens.

For PPO training, 3,505 unannotated log samples with lengths ranging from 61 to 205 tokens are used, contributing about 316k training tokens. All statistics are computed using the LLaMA-2-7B-chat tokenizer.

2) Hyperparameter Setting: In this study, the LLaMA2-7b-chat is used as our base model for fine-tuning and the optimal training parameters for the RLHF and SFT models were determined through an iterative process of validation and performance benchmarking. The Low-Rank Adaptation (LoRA) [43] technique was used for fine-tuning throughout the entire training process.

For the SFT baseline, the LoRA attention dimension was set to 8, with a scaling alpha parameter of 32 and a dropout probability of 0.1 applied to the LoRA layers. The task-specific hyperparameters included a batch size of 4, gradient accumulation steps of 4, a learning rate of 1e-4, and a training duration of 3 epochs.

During the reward modeling phase, the batch size was configured to 4, with gradient accumulation steps set to 2 and a learning rate of 1e-5. The model was trained for 2 epochs, resulting in a reward model optimally suited for the task. After verification, this model achieved an accuracy of 95.33% on the test set.

In the PPO phase, we adapted the implementation of relevant algorithms from the Transformer Reinforcement Learning (TRL) framework [39]. The batch size was set to 16, with a learning rate of 1.5e-5 and gradient accumulation steps of 8. The maximum generation length was fixed at 512 tokens. Each batch was trained over four PPO epochs, using one minibatch per epoch. The KL divergence coefficient β was set to 0.1. The model was trained for a total of 108 steps, and the checkpoint achieving the best validation performance was selected for subsequent analysis. All other hyperparameters were kept at their default values.

The device used in this study is an NVIDIA H20 96GB GPU. In the specific setup, both the SFT and reward modeling stages are trained using a single GPU, while in the PPO training stage, dual GPU parallel computing is used to accelerate the training process.

3) Evaluation Metrics: To evaluate the model's performance, we introduced three key metrics: accuracy, interactivity, and richness. For each metric, experts assigned a score ranging from 1 to 10 based on its definition and relevance to the task requirements.

Accuracy: the generated output should accurately reflect the log data, ensuring that analysis and recommendations are correct, feasible, and logically sound. It must avoid irrelevant or contradictory information and align with actual operational needs.

Interactivity: the content should embody interactive design principles, dynamically adjusting to user inputs and contextual changes to provide personalized and practical responses.

Richness: while ensuring that all information is accurate and reliable, the log analysis should be thoroughly detailed, providing as many relevant insights and feasible solutions as possible to support decision-making and operational practicality.

B. Root Cause Localization Task

1) Dataset: We collected 296,134 alarm records from a practical OTN over a period of 40 consecutive days and extracted 4,430 fault-related records for verification. These verification records include 378 single-root cause scenarios, 177 dual-root cause scenarios, and 76 triple-root cause scenarios. During the training phase, 400 fault scenarios were randomly selected from a total of 631, while 50 scenarios were used to evaluate each task during the testing phase. Then we constructed 6,000 CoT training samples for each of the four tasks involved in alarm localization, which were used in the overall model training process.

From a token-level perspective, the datasets used in the SFT and PPO stages exhibit moderate and well-controlled sequence lengths across different tasks. In the SFT stage, Task 1–4 samples consist of instructions (ranging from 45 to 747 tokens), inputs (12–957 tokens), and outputs (116–2,062 tokens), resulting in total training tokens ranging from approximately 68k to 290k and test tokens from 480 to 60k. For PPO training, prompts composed of instructions and inputs have lengths ranging from 185 to 784 tokens, contributing between 1.4M and 2.8M training tokens. All statistics are computed using the LLaMA-3-8B-Instruct tokenizer.

2) Hyperparameter Setting: In this study, the LLaMA3-8b-Instruct is used as our base model for fine-tuning. During the training process, we developed a ReFT-based enhanced model for four alarm tasks: topology relation extraction (Task 1), alarm propagation relation extraction (Task 2), alarm event matrix generation (Task 3), and RCL of fault events (Task 4).

For the SFT pipeline, the LoRA attention dimension was set to 8, with a scaling alpha parameter of 32 and a dropout probability of 0.1 applied to the LoRA layers. The task-specific hyperparameters included a batch size of 4, gradient accumulation steps of 4, a learning rate of 1e-4, and a training duration of 2 epochs.

In the policy optimization stage, all tasks were configured with a batch size of 16 and a maximum generation length of 1024 tokens. Each batch underwent four PPO epochs, using one minibatch per epoch. The remaining hyperparameters were set as follows: Task 1 used 8 gradient accumulation steps, a learning rate of 2e-6, and a β value of 0.1. Task 2 employed 4 gradient accumulation steps, a learning rate of 2e-6, and a β value of 0.03. Task 3 adopted 8 gradient accumulation steps, a higher learning rate of 5e-6, and a β value of 0.15. Task 4 followed the same settings as Task 1, with 8 gradient accumulation steps, a learning rate of 2e-6, and a β value of 0.1. Among them, we evaluated β values of 0.005, 0.01, 0.03, 0.05, 0.1, and 0.2, and observed that the range 0.03–0.2 consistently yields stable performance, among which the final values were selected in combination with other PPO hyperparameters to ensure balanced policy stability and reward dynamics.

The entire model fine-tuning process for the four alarm localization tasks was performed using three NVIDIA H20 96GB GPUs.

3) Evaluation Metrics: To evaluate the performance of the ReFT algorithm on this task, we adopted accuracy as the

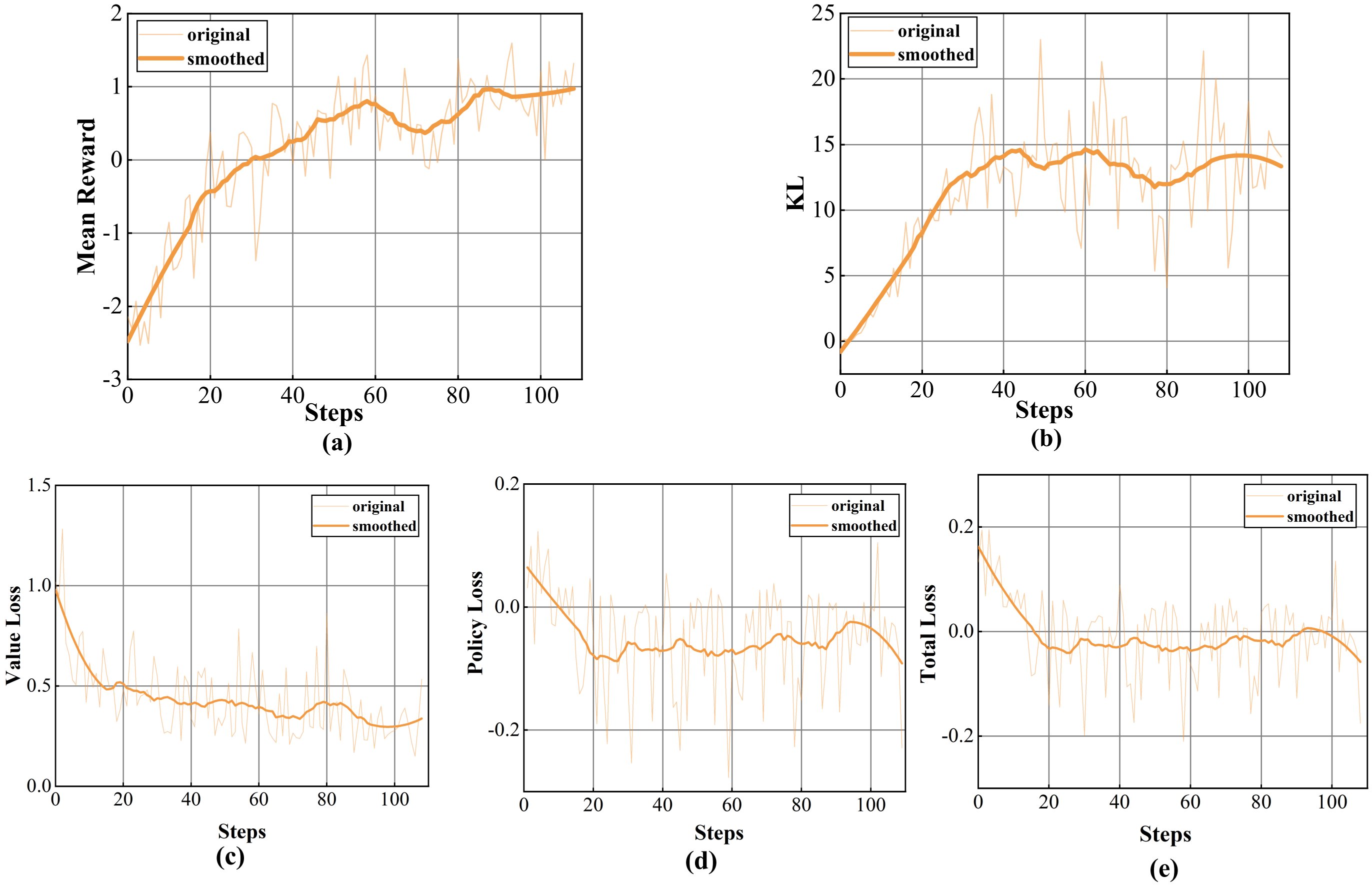

Fig. 5. The results of PPO training in the RLHF-based log analysis task, including (a) the mean reward curve, (b) the KL divergence curve, (c) the value loss curve, (d) the policy loss curve, (e) the total loss curve.

primary evaluation metric. For this indicator, the score will be taken between 0-1.

Accuracy: measured by manually evaluating the consistency between each inference step in the model-generated response for each task and the corresponding reference answer. In this evaluation, any errors in key characters or numerical values are regarded as incorrect results.

VI. Demonstrations and Results

To validate the effectiveness of the proposed methodologies, this section conducts a comprehensive analysis of the performance improvements in log analysis tasks implemented through the RLHF method. In parallel, we evaluate the improvement in the accuracy of CoT alarm tasks achieved by adopting ReFT technology. Additionally, we explore the influence of different reward models and PPO initial policies on the overall performance of RLHF.

A. The Results of the Log Analysis Task Based on RLHF

In this section, we demonstrate the training of log analysis tasks in the RLHF stage and the performance comparison with the training in the SFT stage.

Firstly, the mean reward curve demonstrated a continuous upward trend throughout the entire training process, as shown in Fig. 5 (a). This indicates that the PPO algorithm effectively optimizes the policy, resulting in higher rewards for LLM in the task. The fluctuation of the curve reflects that the model gradually achieves a balance between exploration and exploitation under the guidance of the reward model. In addition, the fluctuation of the KL divergence curve, as shown in Fig. 5 (b), with a peak of about 20, indicates significant differences between the new policy and the previous policy. Given that our task requires a substantial deviation from the initial policy while avoiding pattern collapse, this change in the training curve is expected and considered acceptable.

In addition, to further illustrate the stability of PPO training, we report the evolution of key optimization metrics in Fig. 5(c)–(e), which respectively present the value loss, policy loss, and total loss during PPO updates. Despite the relatively small number of update steps, all three losses remain within bounded and stable ranges throughout training, without exhibiting monotonic increase, divergence, or collapse. Specifically, the value loss gradually decreases and oscillates within a narrow interval, indicating stable fitting of the value function to the advantage estimates. Meanwhile, the policy loss fluctuates symmetrically around zero due to the clipped PPO objective, reflecting normal stochastic policy updates without gradient instability. The total loss remains well balanced, demonstrating effective coordination among policy optimization, value learning, and entropy regularization.

As an enhanced version of the SFT model, the preference model has achieved an accuracy improvement of 1.64 points and a richness increase of 1.08 points compared to the SFT

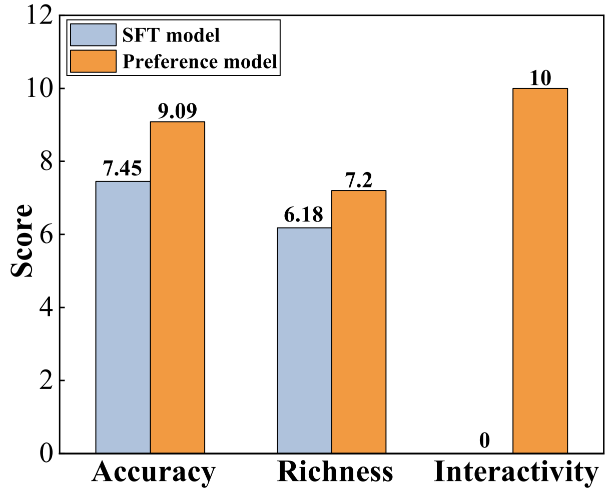

Fig. 6. Performance comparison between the SFT model and the preference model in the RLHF-based log analysis task.

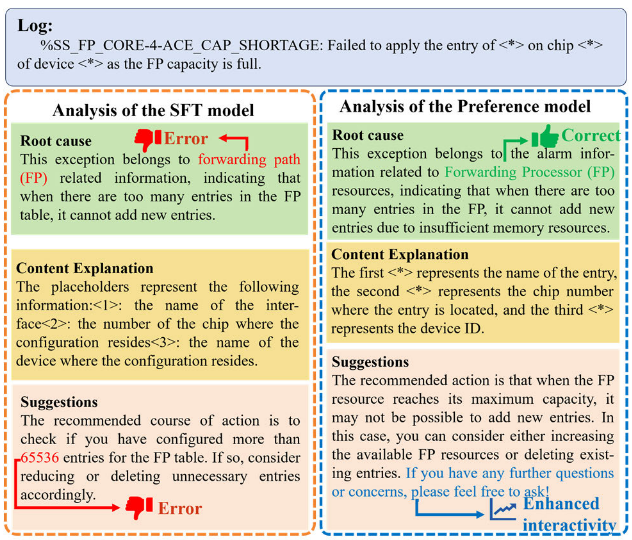

Fig. 7. Response comparison between the SFT model and the RLHF model in the RLHF-based log analysis task.

model. In addition, interactivity showed the most substantial improvement, with an increase of 10 points, underscoring the key role of RLHF training in enhancing user interaction. This enhancement ensures that the model's output aligns more closely with human communication patterns, rather than merely providing static answers, as shown in Fig. 6.

It is worth noting that an inherent trade-off exists between richness and accuracy. The increase in richness means that the generated log analysis outputs provide more detailed background information, potential causes of anomalies, and additional solutions, as shown in Fig. 7. This increase in richness occasionally affects accuracy, as the model might introduce speculative or irrelevant information, thereby affecting the overall reliability of the analysis.

To address this trade-off, RLHF uses reward modeling to balance richness and accuracy. Specifically, during the PPO training phase, the model learns human-annotation preferences to expand the amount of information generated while maintaining the accuracy of key content. The evaluation results

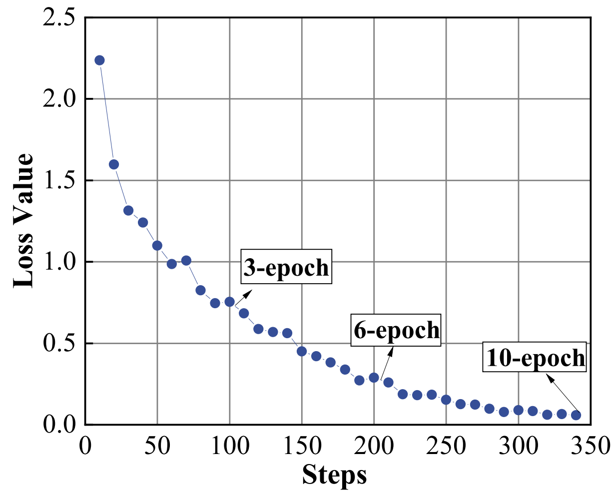

Fig. 8. Training loss curve of the SFT model during the RLHF-based log analysis task.

indicate that although the richness increased by 1.02, it did not significantly affect the overall improvement in accuracy. This discovery indicates that the RLHF mechanism effectively guides the model to generate more comprehensive outputs while maintaining a high level of correctness.

In addition, to better understand these improvements, it is essential to understand the differences between the two training stages. In the log analysis task, the training data in the SFT stage mainly focuses on identifying the root cause of abnormal logs and providing direct solutions. Due to the imitative nature of SFT training, the model outputs tend to closely resemble the original labeled examples. However, by combining preferences for accuracy, richness, and interactivity, the performance of LLM can be further enhanced. In the process of RL, the model effectively integrates human-labeled data, its intrinsic abilities, and human preferences.

In summary, these results indicate that RLHF training improved the accuracy score of LLM in log analysis tasks from 7.45 to 9.09, richness from 6.18 points to 7.2 points, and interactivity from 0 points to 10 points. This demonstrates that RLHF enhanced the alignment between the model's output and human intent while mitigating the hallucination problem.

B. The Influence of Different Initial Policies Based on RLHF

In RLHF, the initial policy selection of the PPO algorithm markedly influences the final model performance. To evaluate its impact, we selected SFT models trained for 3, 6, and 10 epochs, respectively, as the initial policy for PPO under the same training parameters. The corresponding SFT loss curves are presented in Fig. 8 to highlight the differences among these epochs.

Then, under the same reward model and PPO training parameters, the three initial policies exhibited distinct behaviors. The policy trained for 3 epochs exhibited faster reward growth and achieved a higher final return, whereas training for 10 epochs resulted in slower reward improvement and a lower final return, as shown in Fig. 9(a), suggesting potential overfitting with increased training epochs. This pattern aligns with the KL-divergence trend in Fig. 9 (b). The policy with

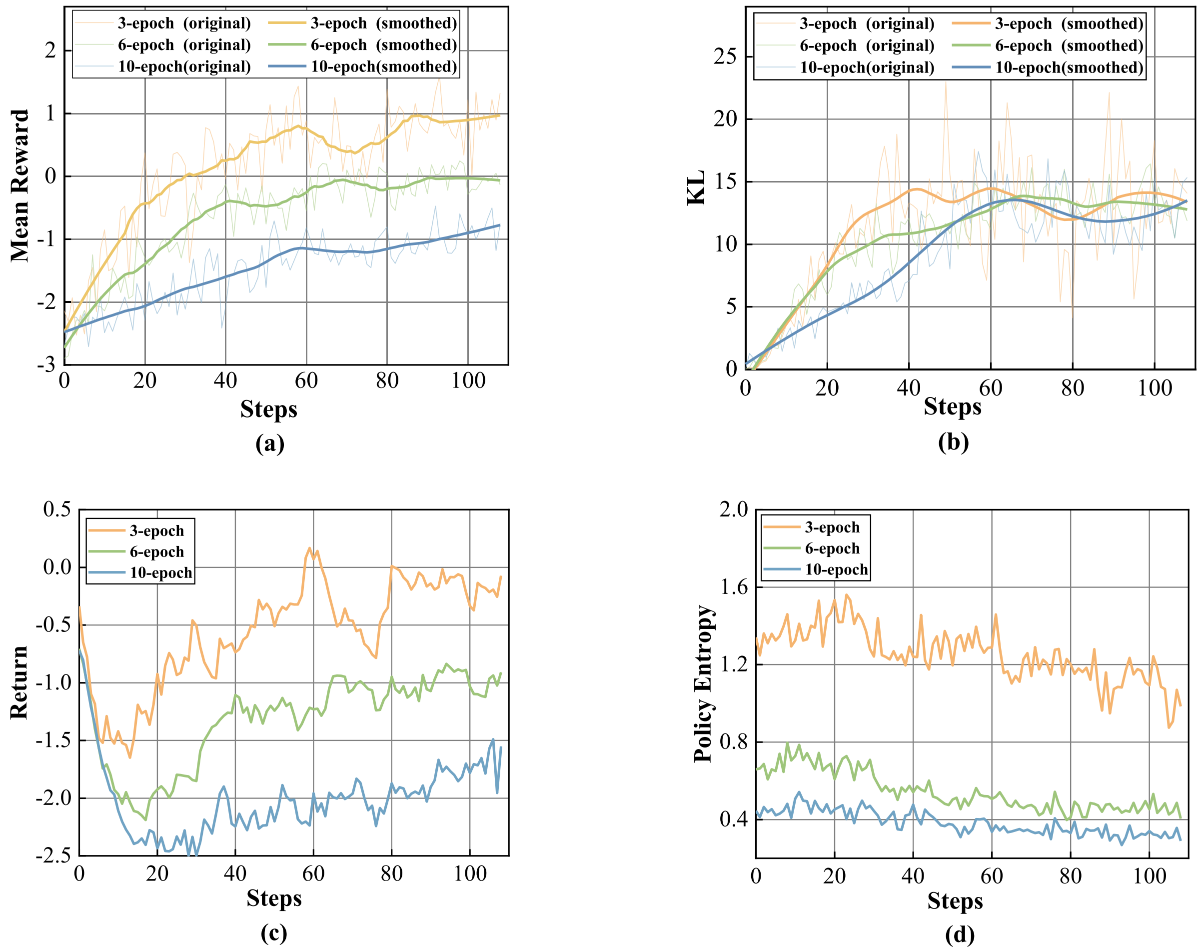

Fig. 9. Comparison of the following metrics during PPO training for log analysis tasks using different initial policies, i.e., SFT models trained for 3, 6, and 10 epochs: (a) mean reward, (b) KL divergence, (c) return, and (d) policy entropy.

an epoch of 3 shows a large and rapid policy adjustment (high KL value), while the policy with an epoch of 10 shows smaller fluctuations, reflecting greater stability.

To verify the above conclusion, we further analyzed the variation curve of policy entropy, as shown in Fig. 9 (d). In theory, entropy values more intuitively reflect the exploratory nature of the policy. Higher entropy values indicate greater output diversity, while a lower entropy value indicates a more fixed output pattern. The results show that the initial policy entropy decreases as the number of SFT epochs increases: it is highest at epoch 3, lower at epoch 6, and lowest at epoch 10. This suggests that extended SFT training leads to overfitting on expert data and reduced exploratory behavior. The return curves in Fig. 9 (c) support this observation: the initial policy at epoch 3 achieves higher returns, while that at epoch 10 struggles to improve.

In addition, we evaluated the performance of different initial policies and preference models trained through PPO on the test set using specified evaluation metrics, as shown in Fig. 10.

We first analyze the performance of SFT models with different epochs on three indicators. The results shows that accuracy improves with the increase of SFT epochs, while richness shows a decreasing trend. This is because accuracy measures the consistency between generated content and logs,

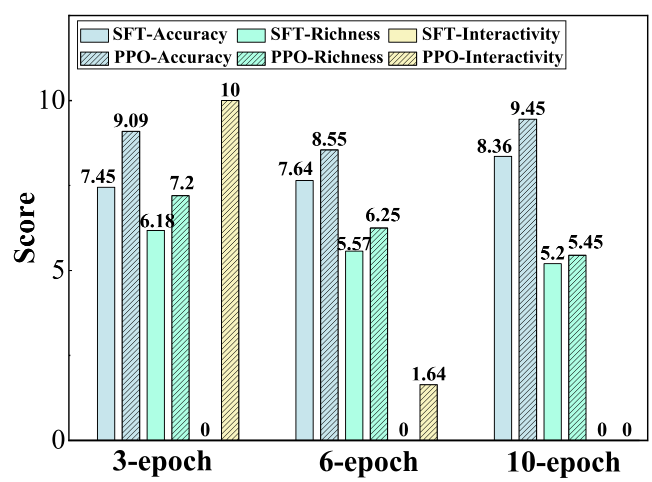

Fig. 10. Performance evaluation results of preference models based on different initial policies (SFT models trained for 3, 6, and 10 epochs) and training stages (SFT and PPO stages).

while richness, to some extent, evaluates the completeness of the log background, reasons, and recommendations. As SFT training deepens, the model tends to improve accuracy at the cost of reducing richness. Because the SFT stage does not incorporate interactivity, the interactivity score for all SFT models is 0.

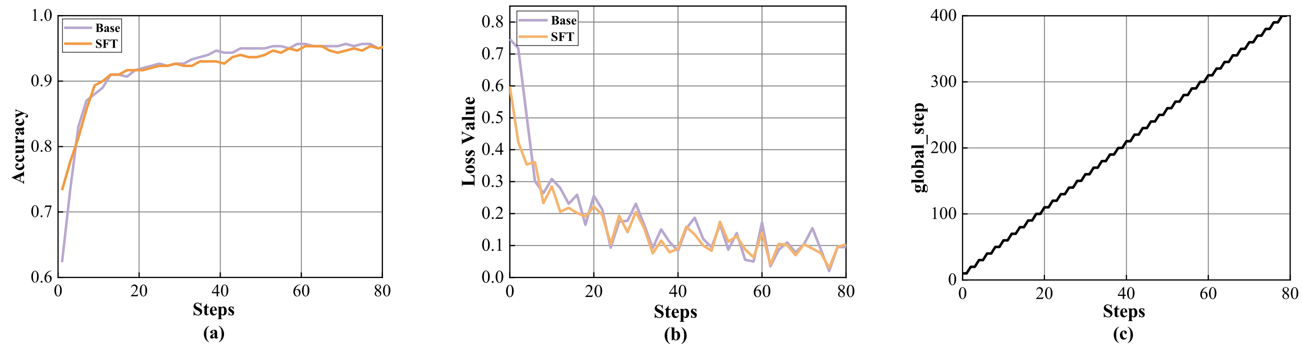

Fig. 11. Training process comparison between initializing the reward model with the base model and initializing the reward model with the SFT model in the RLHF-based log analysis task, including (a) accuracy curve. (b) training loss curve. (c) training global steps curve.

Secondly, by comparing the metrics of content generated by the SFT model and the preference model, we can conclude that regardless of the initial policy epoch of 3, 6, or 10, the PPO-trained preference model outperforms the SFT model in terms of generation quality, validating the effectiveness of the method. However, there are significant differences in the improvement of the different epoch models: when epoch is 3, accuracy increase by 16.4%, richness increases by 10.2%, and interactivity increases by 100%; When epoch is 6, accuracy increased by 9.1%, richness increased by 6.8%, and interactivity increased by 16.4%; When epoch is 10, accuracy increased by 10.9%, richness only increased by 2.5%, and interactivity did not improve. These results further confirm that models trained with fewer epochs possess greater plasticity in PPO training, especially when significant adjustments are needed. In this test, models trained for 3 epochs were able to achieve more significant performance improvements.

To summarize, for domain adaptation, initializing PPO from a lower-epoch SFT checkpoint yields higher RLHF returns than using a higher-epoch SFT model, once basic domain knowledge has been learned. This is because excessive SFT training tends to overfit the model to static annotations, reducing policy entropy and limiting exploration during subsequent PPO optimization, which leads to premature convergence.

C. The Influence of Different Reward Models Based on RLHF

The prediction accuracy of the reward model plays a pivotal role in determining the quality of the reward signal during the PPO iteration process, thereby influencing both the direction and effectiveness of policy updates. Consequently, the performance of the reward model is a key factor in successful preference alignment and in enhancing the overall effectiveness of the PPO algorithm [44].

Moreover, downstream PPO training exhibits smooth KL divergence growth and stable entropy instead of policy collapse, providing indirect evidence that the learned reward signal is not overly specific to the training preferences.

Based on these considerations, we further investigate the selection strategy of the reward model from two perspectives: 1) Impact of Reward Model Initialization on Performance, which compares initializing the reward model from a base model (i.e., LLaMA-2-chat-7B) versus an SFT model; and

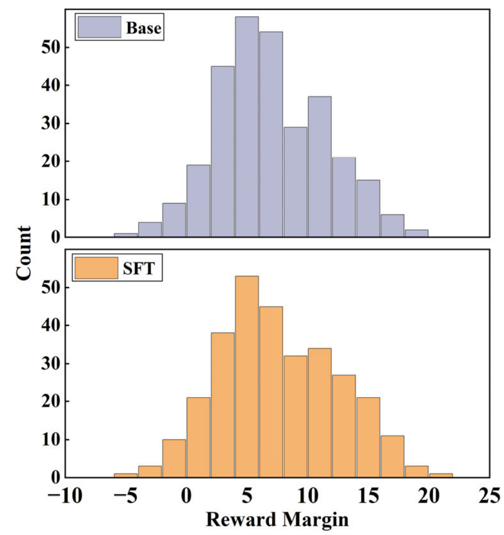

Fig. 12. Comparison of reward margins on the test set for reward models initialized from the SFT model and the Base model in the RLHF-based log analysis task.

2) Epoch Selection for SFT-Based Reward Models, which examines how different training stages of the SFT model affect downstream reward modeling performance.

1) Impact of Reward Model Initialization on Performance: To ensure a fair comparison, we trained both the base and SFT models under identical settings on the same preference dataset and evaluated them at epoch 2. The training loss and validation accuracy curves across 2 epochs are presented in Fig. 11. Results indicate that, benefiting from prior fine-tuning on log analysis tasks, the reward model initialized with the SFT model achieves lower loss and higher accuracy in the early training phase of training. As training progresses, it consistently outperforms the reward model initialized from the base model.

We first evaluated the accuracy of two reward models on the test set, both of which were 95.3%, and presented their margin distributions on the test set, as shown in Fig. 12. The results show that the reward model initialized with the SFT model exhibits a margin distribution concentrated in higher value ranges compared to that initialized with the base model. This indicates that SFT-based initialization improves the model's ability to distinguish between preferred and non-preferred responses.

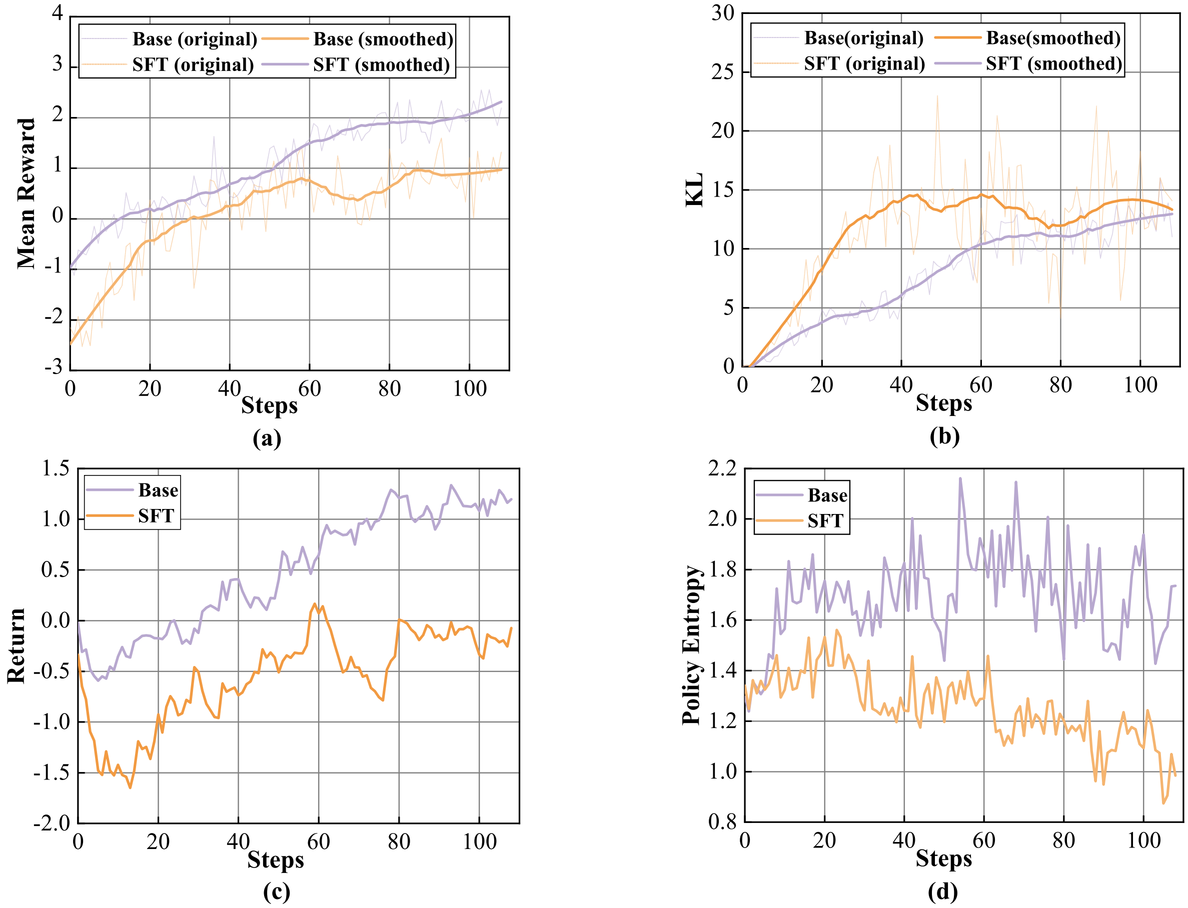

Fig. 13. Comparison of the following metrics for the reward model initialized with the base model versus the SFT model during PPO training in the RLHF-based log analysis task: (a) mean reward, (b) KL divergence, (c) return, and (d) policy entropy.

Under the same PPO initial policy and parameters, the first batch of data is usually generated by the initial policy, so the initial generation is consistent. However, due to the distinct scoring criteria of the two reward models, as shown in Fig. 13 (a). The reward model initialized with the SFT model applies stricter evaluation, leading to lower initial rewards. Combined with the margin distribution findings, this further confirms the base-initialized reward model's weaker discrimination. The KL divergence curve is steadier with the base-initialized model, while the SFT-initialized model triggers sharper adjustments, indicating more active policy updates.

In addition, in the evaluation of the policy entropy curve, as shown in Fig. 13 (d), it can be observed that under the guidance of the reward model initialized with the base model, the overall policy entropy shows a certain upward trend, while under the guidance of the reward model initialized with the SFT model, the policy entropy continues to decrease. This phenomenon indicates that:

The reward model initialized with the base model demonstrates a stronger capacity for exploration guidance. In the PPO training process, the observed increase in policy entropy indicates that the policy distribution gradually becomes more uniform, and the model retains a high degree of uncertainty for various behaviors during the generation process, continuously exploring the policy space. This phenomenon indicates that the reward signal provided by the reward model initialized by the base model is more diverse, failing to converge the policy too quickly and instead encouraging diversity in the policy.

The reward model initialized with SFT model tends to facilitate rapid policy convergence. The continuous decrease in policy entropy means that the model gradually reduces its exploration of different generative policies and begins to focus on certain specific patterns or fixed expressions. This is related to the language pattern preference introduced by the reward model initialized with SFT during the SFT stage, which causes the PPO algorithm to converge more quickly toward deterministic decisions under the guidance of SFT reward signals, narrowing the strategy space and decreasing exploratory value.

However, based on the analysis of the final generated content, although the reward model initialized with the base model maintained a high policy entropy and obtained higher mean rewards (as shown in Fig. 13 (a)) and returns (as shown in Fig. 13 (d)), a pattern preference ending in ellipsis appeared, as shown in Fig. 14, indicating that under the guidance of the reward model initialized with the base model, a higher risk of reward hacking [45], exchanging low-quality exploration and generation for higher rewards. Under the guidance of the reward model modeled by SFT, policies enable more accurate and targeted policy adjustments to better suit human preferences, as shown in Fig. 15.

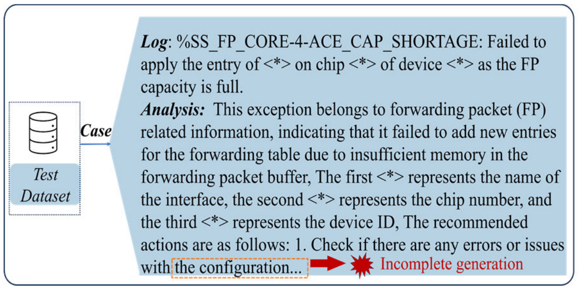

Fig. 14. The final content generated by the preference model guided by the reward model initialized with the base model in the RLHF-based log analysis task.

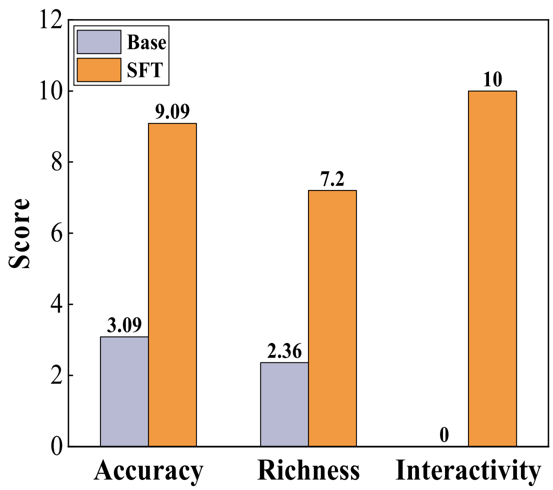

Fig. 15. Comparison of the performance indicators of the final preference models guided by reward models initialized from the SFT model and the Base model in the RLHF-based log analysis task.

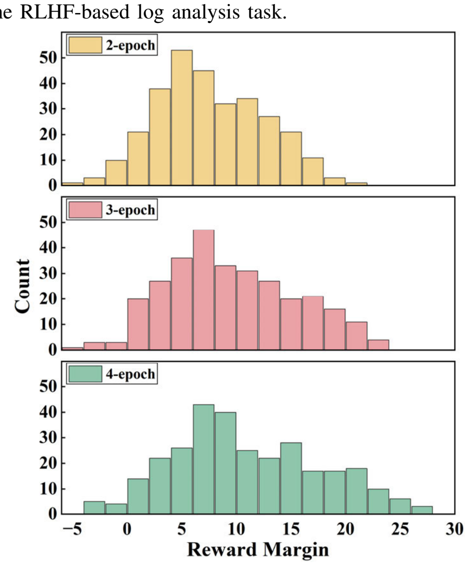

Fig. 16. Comparison of the reward margins on the test set for reward models trained for different epochs (i.e., 2, 3, and 4 epochs) in the RLHF-based log analysis task.

The reward model initialized with the SFT model is more able to clarify the direction of human preference changes compared to the reward model initialized with the Base model, and can accurately guide during the PPO training process.

2) Epoch Selection for SFT-Based Reward Models: This section investigates the impact of reward models at different stages of training on the final preference model, with the SFT model used as the initial reward model. The training loss curve of the specific reward model is shown in Fig. 11 (b). This study selects reward models with training epochs of 2, 3, and 4 for comparison, aiming to analyze the impact of training depth on final performance. First, the reward margin metric on the test set was used to compare the reward models from different epochs in order to evaluate their discriminative ability. The results are shown in Fig. 16.

We observe that as the training deepens, the reward margin distribution on the test set gradually shifts toward higher values, indicating an increasing preference for high-quality samples and a trend toward stable, consistent scoring, as shown in Fig. 17. Additionally, the reward models trained for 2, 3, and 4 epochs achieve test set accuracies of 95.3%, 97.7%, and 97%, respectively.

Then, using the same PPO initial policy and PPO training parameters, we obtained the PPO training curves, as shown in Fig. 17. In the mean reward curve, the reward models at epoch 2 and epoch 3 exhibit high consistency in guiding PPO policy updates. Their initial and final convergence values are relatively close. Combined with the evaluation of the final generated metrics (seen in Fig. 18), this indicates that their preferences are largely consistent.

In addition, the reward model at epoch 4 starts from a lower initial value, indicating a stronger ability to differentiate low-quality text, as shown in Fig. 17 (a). In early training (before 80 steps), the reward curves of PPO policies guided by all reward models were basically consistent. However, at approximately 80 steps, the reward from the epoch-4 model increased sharply, accompanied by a sharp increase in KL divergence and a sharp decrease in policy entropy. The policy gradually converged to a fixed-template generation mode, as shown in Fig. 19, indicating that the reward model at epoch 4 was influenced by more extreme optimization objectives.

Moreover, during the evolution of policy entropy, although there are fluctuations in the three curves, they all exhibit an overall downward trend, which is consistent with the training stability goal of PPO. It is worth noting that in the policy entropy curve, the final entropy value under the epoch 2 configuration always falls between those of epoch 3 and epoch 4, indicating a non-monotonic pattern. Based on the evaluation of the final generated indicators, as shown in Fig. 18, we can conclude that under the guidance of the reward model at epoch 3, the improvement in richness is better than that of the reward model at epoch 2. The optimization objectives during PPO training are more diverse, tending to generate longer and more varied content. As a result, the final entropy is slightly higher than that under epoch 2. However, in the epoch 4 configuration, all responses in the test set ended with "The first" and failed to generate complete outputs, resulting in a lower final entropy value. Due to this result, the specific evaluation of the preference model guided by the reward model is not shown in Fig. 18.

Overall, as the training level of the reward model deepens, the reward signal may become more comprehensive, but it may also bring more extreme update signals. Therefore, the selection of specific rounds still needs to be balanced based on the evaluation indicators of different tasks.

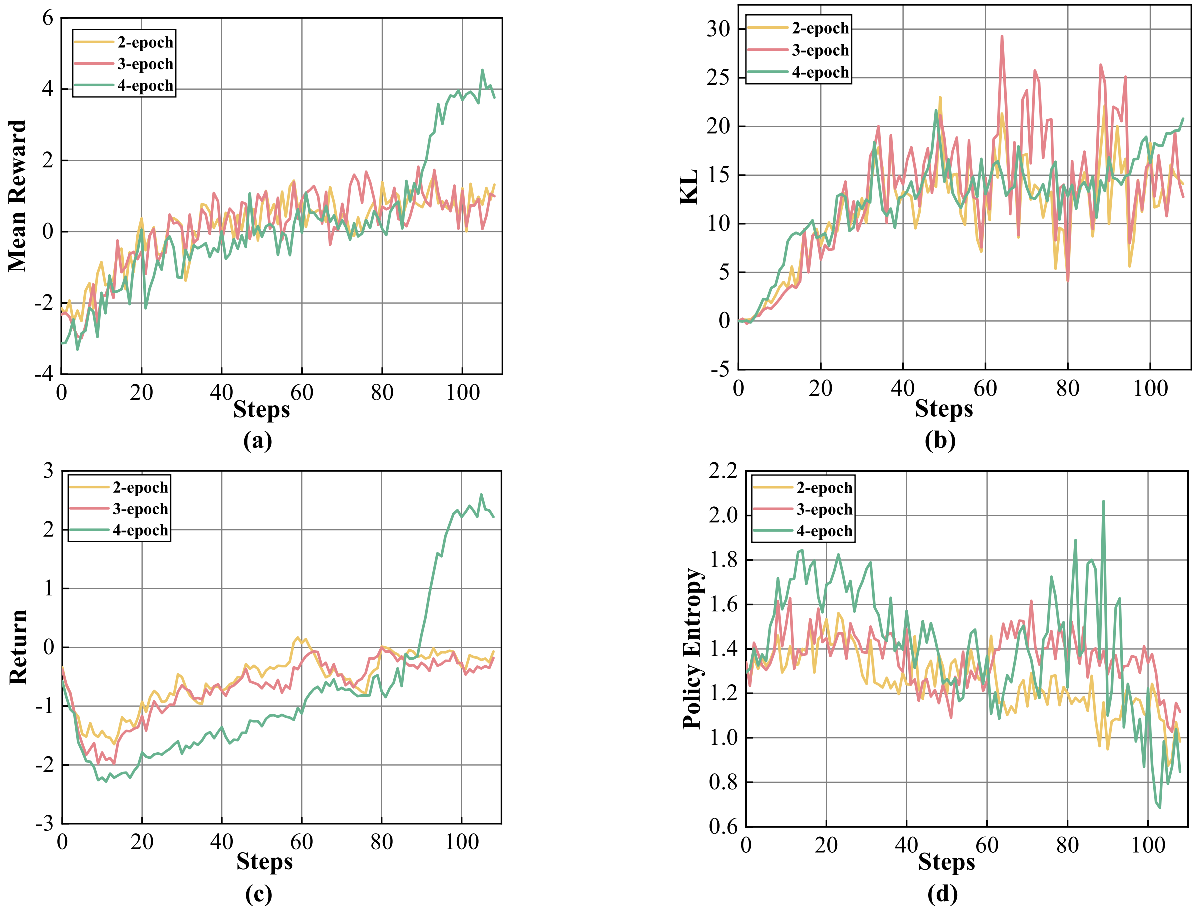

Fig. 17. Comparison of PPO training performance under reward models from different training epochs (i.e., 2, 3, and 4 epochs) in the RLHF-based log analysis task, showing: (a) mean reward, (b) KL divergence, (c) return, and (d) policy entropy curves.

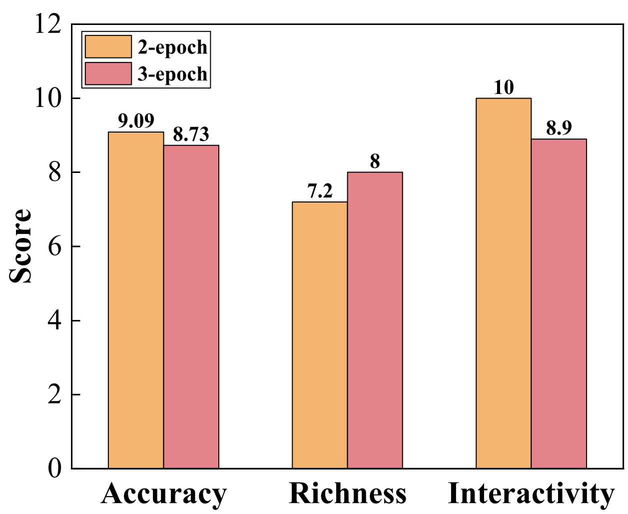

Fig. 18. Comparison of the performance of the final preference models guided by reward models at different epochs (i.e., 2, 3, and 4 epochs) in the RLHF-based log analysis task.

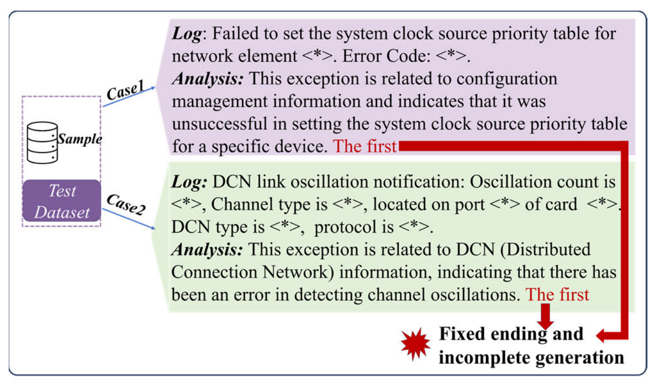

Fig. 19. The final performance of the preference model guided by the reward model at epoch 4 in the RLHF-based log analysis task.

D. Evaluation of SFT and RLHF on LLaMA-3-8B-Instruct

In the original analysis, LLaMA-2-7B-chat-hf was adopted as the base model to construct the SFT and RL framework. To further evaluate the generality of the proposed method on stronger foundation models, additional analyses were conducted using LLaMA-3-8B-Instruct as the base model. In these analyses, the same preference dataset and reward model used in the previous analysis were retained to ensure a fair comparison.

Specifically, PPO-based reinforcement learning optimization was performed on the LLaMA-3 model using the constructed preference model as the reward signal. The training dynamics are illustrated in Fig. 20, which shows the evolution of the KL divergence and the average reward score during training. As shown in the figure, the mean reward score increases steadily from negative values in the early stage to stable positive values in later iterations, indicating

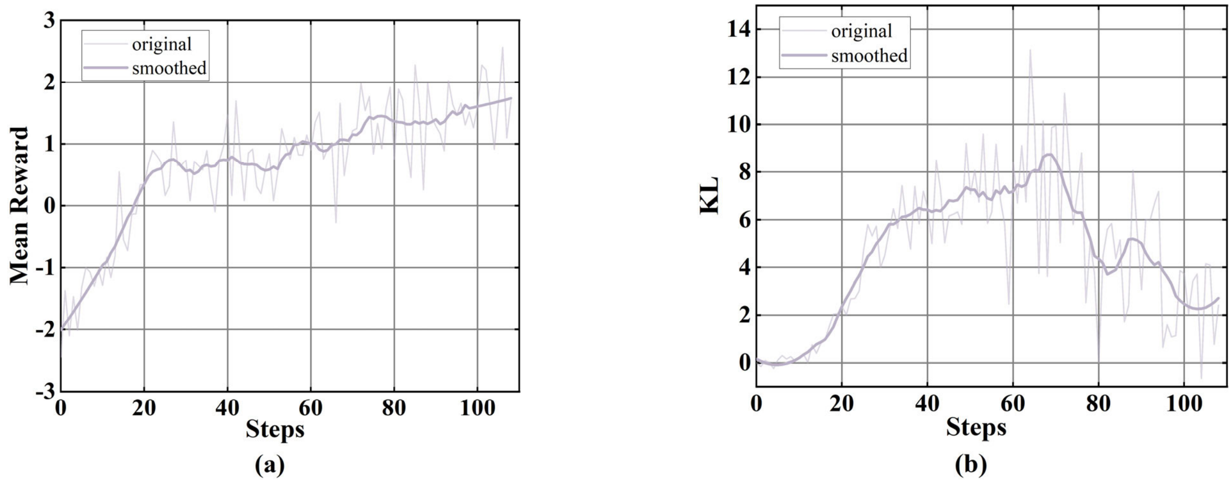

Fig. 20. RLHF training dynamics with LLaMA-3: (a) mean reward curve; (b) KL divergence curve.

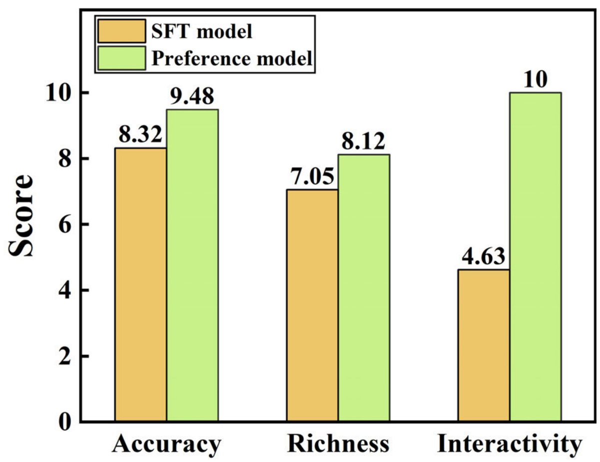

Fig. 21. Performance comparison between the LLaMA-SFT model and the preference model.

that the policy gradually learns to generate responses that better align with the preference model. Meanwhile, the KL divergence remains within a controllable range throughout the training process without exhibiting abrupt divergence or collapse. Although moderate fluctuations occur due to the stochastic nature of policy optimization, the overall trend remains stable, demonstrating that the PPO training process maintains a balanced trade-off between reward maximization and policy constraint.

After RL optimization, the model was evaluated using three qualitative metrics, namely Accuracy, Richness, and Interactivity, which assess the correctness of diagnostic conclusions, the completeness of reasoning content, and the ability to provide interactive explanations. The evaluation results are shown in Fig. 21. The LLaMA-3-based SFT model achieves scores of 8.32, 7.05, and 4.63 on Accuracy, Richness, and Interactivity, respectively. After applying preference-based reinforcement learning, the optimized model improves to 9.48, 8.12, and 10.00 on the corresponding metrics. The corresponding results using LLaMA-2 as the base model, shown in Fig. 6, are slightly lower across all three metrics. These results suggest that a stronger base model can improve diagnostic accuracy, while RL further enhances reasoning richness and interactivity.

In addition, a representative generation example is presented in Fig. 22 to illustrate the qualitative improvements introduced by the preference optimization. Compared with the SFT model, the RLHF-optimized model provides more precise explanations for logging events and clearer guidance for troubleshooting. While the SFT output only offers a general description of RSPAN (Remote Switched Port Analyzer) or mirror configuration issues and suggests checking the configuration, the RLHF output specifies potential causes, such as incorrect mirror settings or software bugs. It also provides actionable steps, including verifying and reconfiguring the mirror session, upgrading firmware if necessary. Furthermore, the RLHF output encourages follow-up interaction, which aligns better with human preferences.

In summary, compared with LLaMA-2, LLaMA-3 achieves higher scores across all evaluation metrics, confirming that a stronger base model generally improves performance. However, additional studies show that, despite its strong general language capabilities, LLaMA-3 still faces domain-specific challenges, such as accurately handling specialized knowledge and reasoning within the optical network domain. Applying RLHF in this context still effectively enhances the model's alignment with the preference dataset and improves task-specific reasoning performance.

E. The Results of the Alarm Task Based on ReFT

In this section, we evaluate the effectiveness of the proposed method across four distinct alarm tasks by independently training a model for each of the four tasks and recording the variations in mean reward and KL divergence during training, as shown in Fig. 23.

Overall, the results demonstrate consistent and stable training behavior across all tasks. In Task 1, the mean reward showed noticeable fluctuations in the early phase but gradually increased and stabilized around 0.8, while the KL divergence first rose and then declined, indicating active exploration followed by convergence to a stable policy. In Task 2, the mean reward remained at a relatively high and stable level throughout training, suggesting that the model was able to acquire an effective policy early on. The KL divergence in

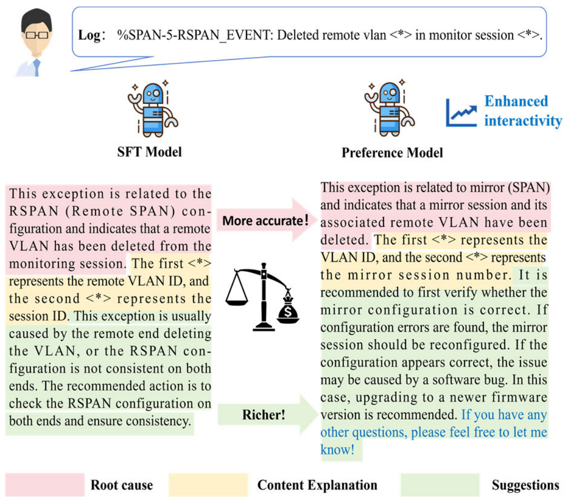

Fig. 22. Comparison of generated responses of the SFT model and the preference-optimized model.

this task increased moderately during the initial phase, reached a peak around mid-training, and subsequently decreased to a stable level, reflecting effective control over policy shifts.

Similar trends were observed for tasks 3 and 4. In Task 3, the mean reward steadily increased with slight fluctuations before stabilizing above 0.9, while the KL divergence fluctuated greatly throughout the training process and eventually stabilized at a lower value. In Task 4, the mean reward also showed a gradual upward trend, with fluctuations in early training gradually smoothing out. The KL divergence followed a pattern of initial increase, reaching a mid-training peak before decreasing to a lower, stable level. These results collectively suggest that the proposed method can maintain a balance between reward maximization and policy stability across varied task settings, validating its generalization capability and robustness.

In addition, we introduce accuracy indicators to evaluate the performance of domain-adapted LLM in alarm analysis tasks. Considering the strict requirements for fault tolerance in alarm tasks, only when the model output is completely consistent with the ground truth, it is considered correct during manual evaluation. In this evaluation, any errors in key characters or numerical values are regarded as incorrect results.

The accuracy for all four tasks improves steadily across the two training stages, as shown in Fig. 24. The SFT model achieved an accuracy rate of 86% to 96% on four tasks, but there is still an accuracy bottleneck. The ReFT can further improve performance for all four tasks (accuracies >96%), especially in task 2, extracting fault event alarm propagation relationships, and task 3, generating fault event adjacency matrices, with an accuracy rate of 99%, providing strong support for the landing application of LLMs in alarm analysis scenarios.

In terms of improvements in content generation, since the tasks share common issues and enhancements, we take Task 4 as a representative case for detailed analysis. In the SFT stage, the response of the model to alarm analysis tasks mainly manifests as two problems: incomplete content generation, where some responses only include partial reasoning steps or analysis conclusions, and fail to fully cover the entire reasoning process and final conclusions; the reasoning errors in the reasoning steps and processes, such as incorrect logical deduction, matrix generation, and incorrect RCL results, as shown in Fig. 25. In contrast, after entering the ReFT stage, the responses of model not only achieved more complete and coherent content generation, but also effectively corrected logical errors within the reasoning process, even when supervision was applied solely to the final results. This ultimately led to a significant improvement in the overall accuracy and reliability of the model's responses. This phenomenon can be attributed to the implicit alignment effect of RLHF, where models are guided to generate reasoning chains that are more likely to lead to correct final outcomes.

The tasks have been successfully improved through the implementation of ReFT technology, which further indicates that in RL-based fine-tuning, although only the correctness of the final analysis results is directly supervised, the reward function implicitly encourages the generation of logically reasonable and human-preferred intermediate reasoning steps. Through iterative reward-guided optimization, the model gradually learns to generate inference processes that conform to expected patterns, as incorrect intermediate steps often result in lower rewards due to incorrect final outcomes.

In summary, the demonstration results indicate that ReFT training substantially enhanced the performance of the LLM in alarm tasks, with accuracy improvements from 96% to 98% for Task 1, 86% to 99% for Task 2, 94% to 99% for Task 3, and 90% to 96% for Task 4. These findings validate the effectiveness of the ReFT-enhanced LLM for alarm analysis and demonstrate its promising potential for the development of specialized LLMs in future automatic optical network systems.

F. ReFT Training Based on Multi-Task Mixture

Although the above studies design task-specific reward functions for individual tasks, this does not imply that the proposed approach is limited to single-task fine-tuning. In fact, the same framework can be naturally extended to multi-task scenarios. To demonstrate this capability, we conduct an additional study in which the four subtasks involved in the root alarm localization problem are jointly incorporated into a unified reinforcement fine-tuning process.

Specifically, during the SFT stage, data from the four related tasks are combined to form a joint training set for supervised fine-tuning. During reward construction, task-specific reward functions are integrated into a unified reward framework, with the applied reward determined by the task context of each sample. In the reinforcement fine-tuning stage, rather than fully merging PPO datasets across tasks, we sample 6,000 mixed instances according to a fixed ratio of Task 1: Task 2: Task 3: Task 4 = 1:1:1:3. The total number of training instances is fixed to 6,000 to balance training stability and computational efficiency, while ensuring sufficient task diversity for effective reinforcement learning.

The training dynamics of the mixed multi-task ReFT process are illustrated, as shown in Fig. 26. The mean reward exhibits a clear upward trend and gradually stabilizes after

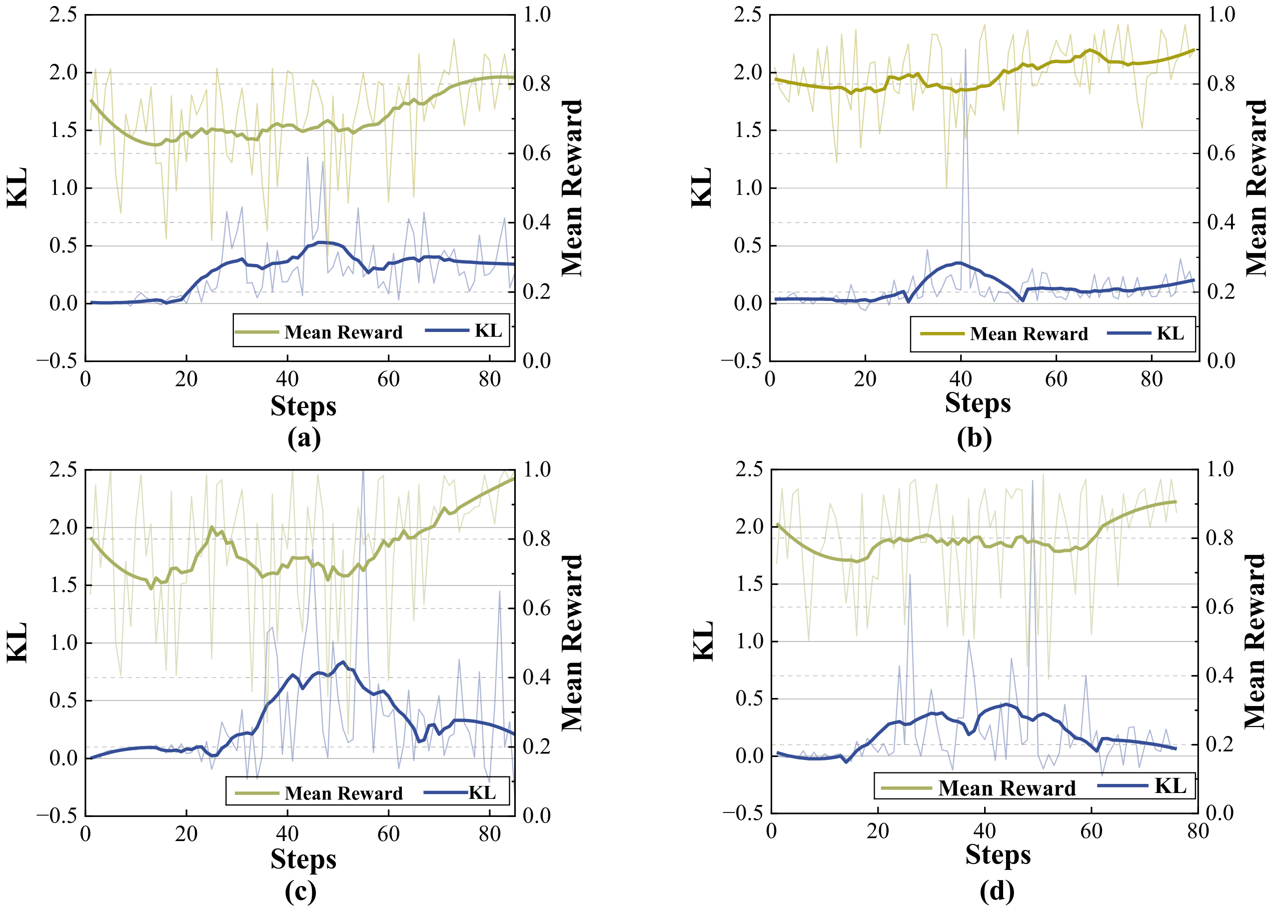

Fig. 23. Mean reward curve and KL divergence curve during the PPO optimization process for (a) Task-1, (b) Task-2, (c) Task-3, and (d) Task-4 in the ReFT-based alarm task.

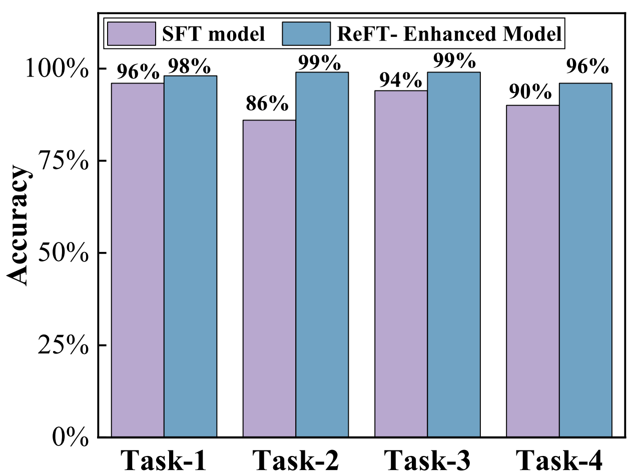

Fig. 24. Comparison of accuracy between the SFT model and the ReFT-enhanced model.

approximately 60 steps, indicating that the policy consistently aligns better with the integrated reward objectives across different tasks, as shown in Fig. 26 (a). Despite the inherent heterogeneity among subtasks, no oscillatory or divergent behavior is observed, suggesting that the mixed-task training strategy remai ns stable under the proposed sampling scheme.

Meanwhile, the KL divergence between the updated policy and the reference model decreases steadily and remains within a controlled range throughout training as shown in Fig. 26 (b). This behavior implies that the PPO updates effectively balance exploration and policy regularization, preventing excessive deviation from the pretrained distribution while still enabling meaningful policy improvement. The smooth decay of KL divergence also indicates the absence of policy collapse, even when multiple task-specific rewards are jointly optimized.

In addition, it can be observed that the used ReFT framework consistently improves performance across all four subtasks compared with the SFT-only baseline as shown in Fig. 27. Specifically, ReFT yields absolute accuracy gains of 2%, 6%, 13%, and 6% on Task 1–Task 4, respectively. Notably, the improvement is more pronounced on Task 3 and Task 2, which involve higher reasoning complexity and stronger dependency on contextual alarm semantics. This indicates that the unified reward-driven optimization can effectively enhance the model's decision boundaries beyond what is achievable through supervised learning alone.

Overall, these results demonstrate that the proposed framework can successfully generalize from single-task to multi-task reinforcement fine-tuning. By integrating task-aware reward functions and carefully controlling the mixed-task sampling ratio, the model not only achieves consistent accuracy improvements across all subtasks but also maintains stable and interpretable training dynamics. This confirms the practicality of the proposed approach for complex real-world alarm analysis scenarios involving multiple interrelated tasks.

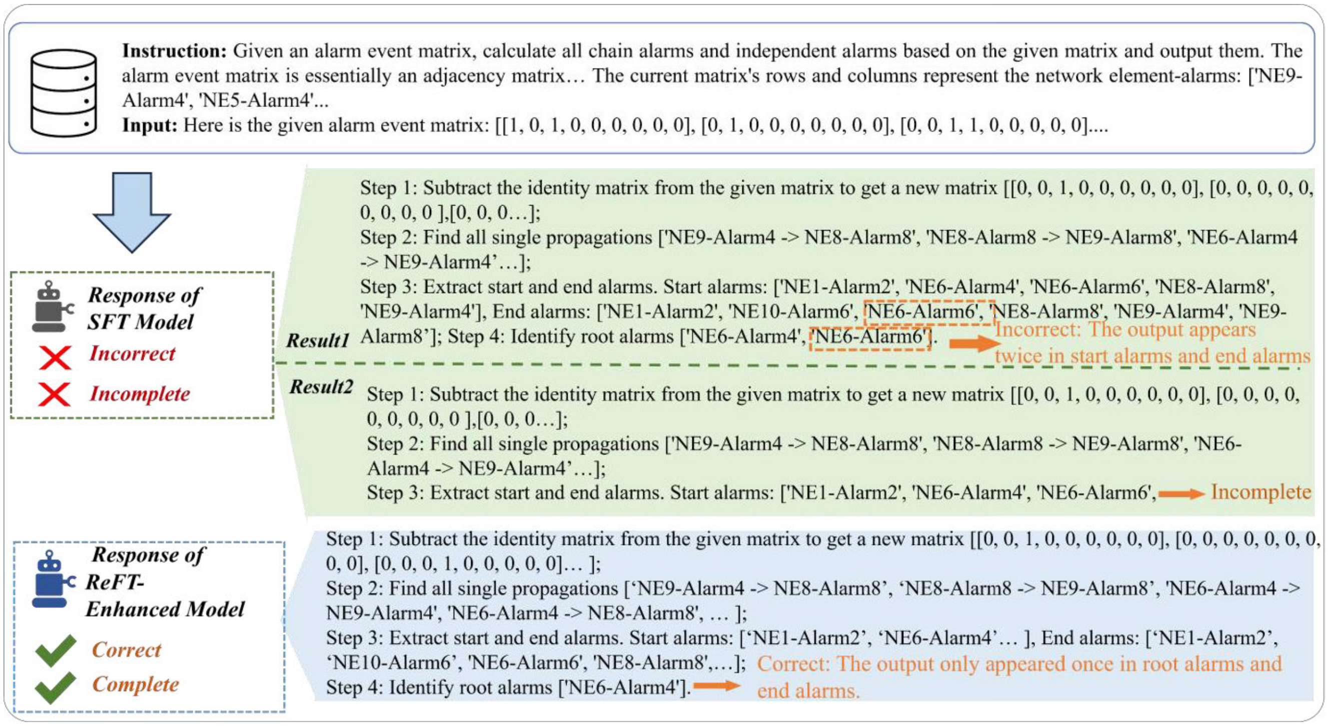

Fig. 25. Performance improvements of the ReFT enhanced model compared with the SFT model, illustrated using Task 4 as an example.

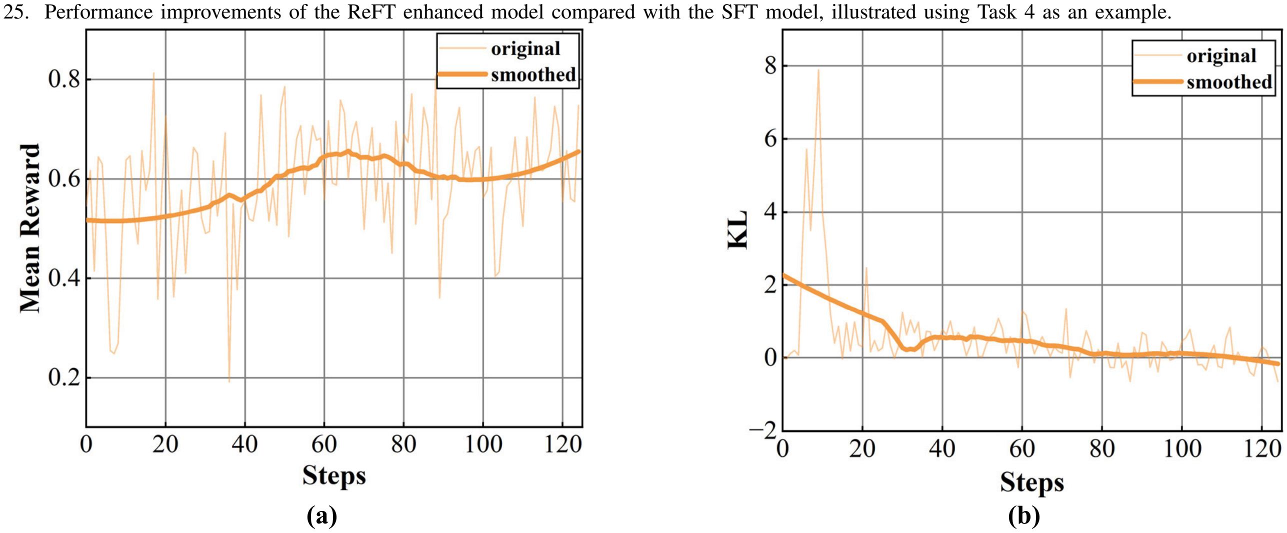

Fig. 26. Training process of Mixed Multi-Task ReFT: (a) the mean reward curve, (b) the KL divergence curve.

G. Retrieval-Augmented Baselines and Task-Specific Performance Analysis

To quantitatively evaluate the effectiveness of RL-based fine-tuning, we compare the proposed RLHF–ReFT framework with strong non-RL baselines on two representative tasks: log analysis and alarm localization. All models are evaluated using the same backbone LLM, identical training and test splits, and consistent inference settings to ensure a fair comparison. The baselines include supervised fine-tuning with carefully designed task-specific prompts (SFT), supervised fine-tuning augmented with retrieval-augmented generation [20] (SFT+RAG), and the proposed RLHF–ReFT framework. For the RAG-based baseline, semantically similar historical cases are retrieved and appended to the input context without introducing explicit causal or structural constraints.

1) Log Analysis Task: Prompt-engineered SFT already provides a strong baseline for the log analysis task, indicating that careful prompt design can effectively guide the model toward correct interpretation of log semantics, as shown in Fig. 28. Incorporating retrieval further improves factual accuracy, as SFT+RAG benefits from direct exposure to relevant historical cases.

However, both SFT and SFT+RAG exhibit limited improvements in response richness and interactivity. This suggests that while supervised objectives and retrieval mechanisms can enhance surface-level correctness, they are less effective in consistently enforcing structured reasoning processes and professional diagnostic behaviors. In contrast, the RLHF-based model achieves the best performance across all evaluation metrics. By explicitly optimizing task-level reward signals

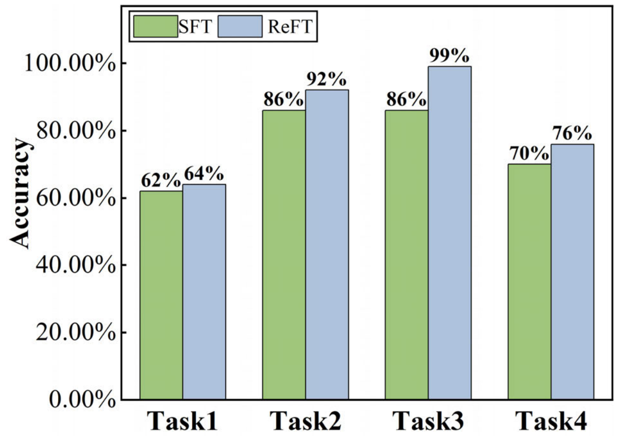

Fig. 27. Comparison of Accuracy Between the SFT Model and the ReFT-Enhanced Model During Mixed Training.

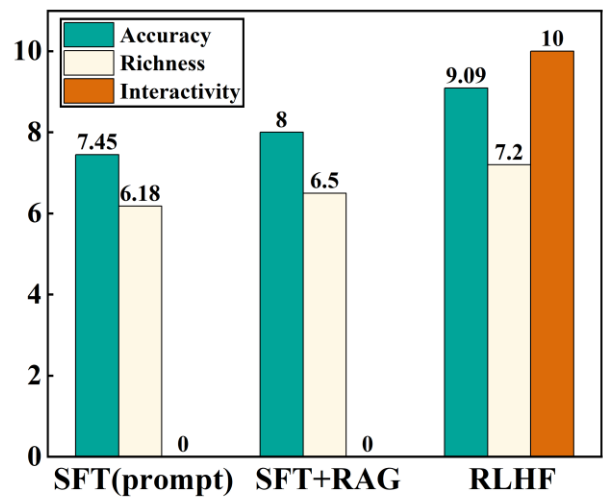

Fig. 28. Comparison of SFT, RAG, and RLHF Results of Log task.

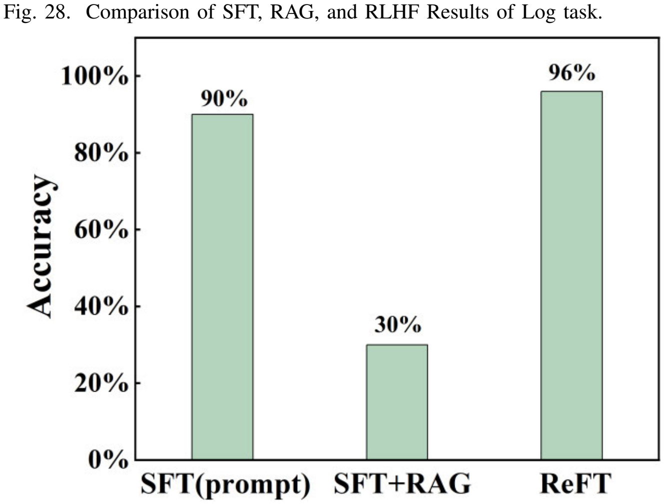

Fig. 29. Comparison of SFT, RAG, and ReFT Results of Alarm task.

that encode correctness, reasoning quality, and response style, RLHF–ReFT enables more effective alignment with domain-specific decision criteria, resulting in more informative and interactive outputs.

2) Root Cause Localization Task: A different and more revealing trend is observed in the alarm localization task, as shown in Fig. 29. While the SFT (Prompt) model attains high root cause localization accuracy, the SFT+RAG baseline suffers a substantial performance drop, despite retrieving semantically similar alarm cases. This counterintuitive result indicates that naïve retrieval based on semantic similarity may introduce misleading correlations, causing the model to confuse root alarms with downstream or chain alarms.

Cause localization inherently requires reasoning over causal propagation and structural dependencies, rather than surface-level semantic similarity. Retrieved examples that are textually similar but structurally irrelevant may bias the model toward incorrect inference paths. In contrast, the proposed RLHF–ReFT framework achieves the highest accuracy, suggesting that RL-based fine-tuning is better suited for capturing causal and propagation relationships by directly optimizing task-specific correctness signals.

Importantly, these results do not suggest that retrieval-augmented generation is ineffective in general. Instead, they highlight that RAG is not well-suited for alarm localization tasks when retrieval lacks causal or topological constraints, whereas RL-based fine-tuning provides a more appropriate alignment mechanism for such reasoning-intensive scenarios.

Overall, these quantitative results demonstrate that prompt engineering and RAG can serve as useful auxiliary techniques, particularly for log analysis tasks. However, they are insufficient on their own for aligning model behavior with domain-specific reasoning and decision-making requirements across tasks. By explicitly optimizing task-level reward signals, RL-based fine-tuning encourages valid reasoning paths while suppressing spurious patterns introduced by retrieved cases, leading to more robust and reliable performance.

VII. Discussion and Future Work

Although the results demonstrate the effectiveness and engineering feasibility of RL-based fine-tuning in optical network O&M scenarios, it is also important to situate these findings within the broader landscape of existing AI-based O&M solutions and to discuss remaining limitations and future research directions.

Traditional AI-based O&M solutions, such as knowledge graph systems and deep learning classifiers, typically rely on predefined rules, structured schemas, or fixed feature representations [46]. While these methods perform well on common and well-defined fault patterns, they often struggle to generalize to heterogeneous log formats and long-tail or previously unseen faults, which are prevalent in real-world optical network environments. In contrast, LLM-based approaches can directly process raw or semi-structured logs and perform semantic understanding and reasoning [21], [47], enabling more robust fault diagnosis and root-cause analysis under complex and uncertain conditions. This fundamental difference partly explains the empirical advantages observed in our studies.

Beyond diagnostic accuracy, LLM-based O&M approaches also provide unique benefits in terms of human–machine interaction and system adaptability. By supporting natural language–based, human-in-the-loop analysis, LLMs allow operators to iteratively refine diagnostic results and better align

TABLE I
COMPARISON OF LLMS AND TRADITIONAL O&M METHODS

Dimension	Knowledge Graph (KG)–based Methods	Traditional DL / DRL-based Methods	LLM-based Method (This Work)
Log & Alarm Representation	Structured entities, alarms, and predefined relations	Hand-crafted or learned fixed features (e.g., templates, embeddings)	Raw or semi-structured logs and alarms with semantic understanding
Generalization to Unseen Log/Alarm Patterns	Limited to predefined schemas and causal links	Poor generalization to unseen formats or fault patterns	Strong generalization via pre-trained knowledge and reasoning
Root Cause Localization	Restricted to known root causes encoded in KG	Dependent on labeled fault categories	Supports multi-root-cause and unseen root-cause reasoning
Explainability	Explicit causal paths but rigid	Limited interpretability	Natural-language explanations with step-by-step reasoning
Human–Machine Interaction	Minimal (rule-based dashboards)	Limited (fixed input–output)	Interactive, conversational, and operator-in-the-loop
Adaptation to Network Changes	Manual graph updates required	Retraining with new labeled data required	Prompt-based adaptation; supports few-shot / zero-shot learning
Deployment Flexibility	Low	Moderate	High
Computational Resource Consumption	Low	Medium to High (training-dependent)	High during training; Moderate during inference
Time Cost (Model Adaptation & Update)	High (manual knowledge engineering)	High (retraining and relabeling required)	Low to Medium (prompting or lightweight fine-tuning)
Suitability for Real-world Optical O&M	Effective for head-class, known faults	Effective with sufficient labeled data	Effective across common, rare, and evolving fault scenarios

model behavior with real operational workflows [48]. Moreover, LLMs significantly reduce adaptation costs in dynamic network environments where concept drift and cold-start issues are common by enabling prompt-level adaptation or few-shot learning [49], rather than requiring costly retraining or manual knowledge graph updates. These properties make LLM-based solutions more practical and scalable for evolving optical networks.

Despite these advantages, several limitations remain and motivate future research.

A. Data Dependency and Generalization

The framework relies on high-quality human preference data for RLHF and large-scale structured alarm data for ReFT, both of which require substantial expert involvement for annotation or validation. This data dependency may limit scalability and rapid adaptation in scenarios with scarce labeled data. Future work will explore weakly supervised or semi-automatic preference construction methods, such as leveraging structural log features, historical remediation records, or rule-based systems. In addition, transfer learning and cross-scenario data reuse strategies may further enhance model generalization across heterogeneous network environments.

B. Computational Overhead and Deployment Scalability

RLHF and ReFT introduce higher computational costs than supervised fine-tuning due to reward modeling and PPO-based optimization. While sufficient GPU resources were available in this study, frequent RL-based updates may be impractical in resource-constrained or latency-sensitive deployment environments. Future research may focus on lightweight RL fine-tuning strategies, including reduced PPO update frequency, partial parameter freezing, and localized policy optimization. Combining parameter-efficient fine-tuning (PRFT) with reward distillation or offline RL paradigms may also improve deployment scalability.

C. Task Coverage and Method Generality

The proposed framework is validated on two representative tasks, log analysis and alar root cause localization, with task-specific reward designs and reasoning workflows. Extending the framework to more complex O&M scenarios, such as QoT estimation or multi-root-cause fault analysis [50], may require additional reward and hyperparameter tuning. Future work will investigate task-aware and hierarchical reward designs, as well as multi-task RL fine-tuning strategies, to improve adaptability and generality across diverse optical network tasks.

To help readers quickly understand the strengths and weaknesses of different approaches, we have summarized the above analysis in Table I. This table presents the applicability, performance, and limitations of traditional AI-based O&M methods and RL-based fine-tuning approaches in optical network log analysis and alarm localization tasks, providing an intuitive reference for subsequent discussion and method selection.

Beyond the limitations discussed above, reinforcement learning techniques for large language models are evolving rapidly. In this work, reinforcement fine-tuning is implemented using PPO, as it has been widely adopted in early RLHF and ReFT frameworks and serves as a standard optimization approach in LLM alignment studies. This choice ensures methodological consistency with prior work. Nevertheless, recent studies have also proposed new reinforcement optimization methods, such as Group Relative Policy Optimization (GRPO) [51]. Future work will investigate the applicability of these emerging methods in optical network O&M scenarios, as well as explore more scalable reward construction mechanisms to further enhance the practicality of LLM-based O&M systems.

VIII. Conclusion

This paper proposes a domain-adaptation methodology in which existing RLHF and ReFT techniques are employed to fine-tune LLMs, enabling them to perform optical network O&M tasks effectively. Taking log analysis and alarm localization, two key tasks in optical network O&M, as examples, the implementation details of RLHF and ReFT were analyzed, and the final task performance was evaluated. The results showed that after RLHF fine-tuning, the consistency between log analysis tasks and expert intentions was significantly improved, with an accuracy score above 9 and an interactivity score above 10. Although the richness score was slightly lower, it still outperformed that of supervised fine-tuning. For alarm-related tasks, the minimum accuracy also reaches 96%. In addition, we also analyzed and evaluated the selection of initial policies in the PPO stage and the selection of reward models involved in the implementation details of RLHF. These studies were based on specific task requirements and had certain limitations. Future work can include a more comprehensive analysis by incorporating additional evaluation metrics. We hope that this method of strengthening and fine-tuning LLMs can contribute to the professionalization of LLMs in a specialized domain.

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